D Y PATIL AGRICULTURE & TECHNICAL UNIVERSITY TALSANDE

MCA SANDWICH

PROJECT NAME

VISUALIZING HOUSING MARKETS TRENDS: AN ANALYSIS OF SALE PRICES AND FEATURES USING TABLEAU

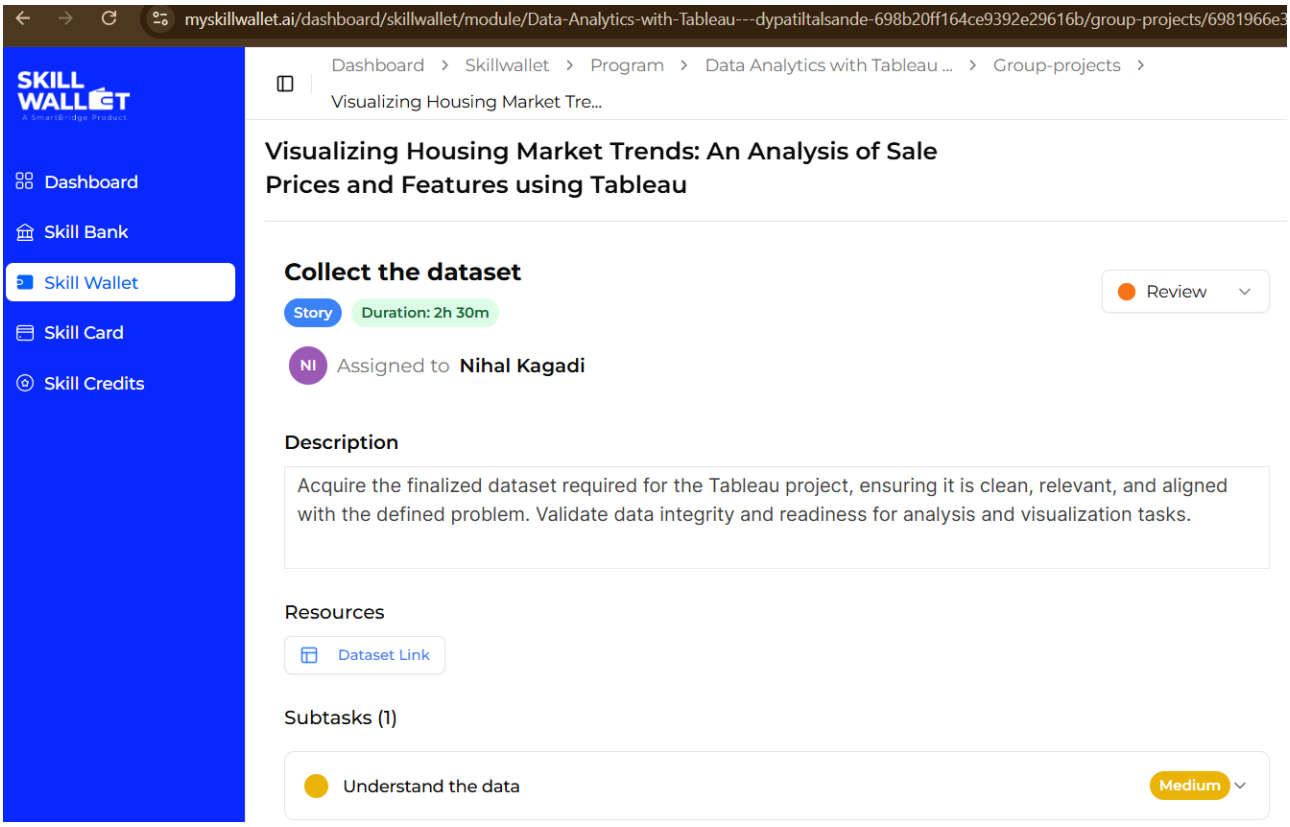
TEAM MEMBERS

1) NIHAL KAGADI
2) DIPALI PATIL
3) SHIVRAJ PATIL

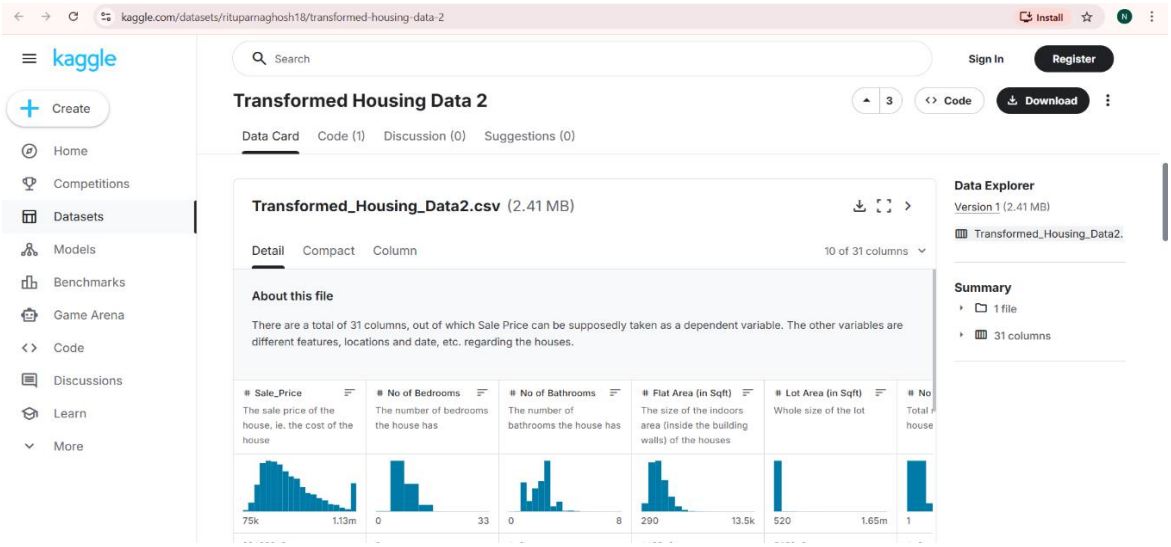
1) Data Collection & Extraction from Database:

❖ Collect the dataset:

1) Data is Collected from Dataset link Which is provided already so click on **Dataset Link**.



2) From the Kaggle Download the Dataset **Transformed_Housing_Data2.csv** by clicking on **Download** button,



❖ Loading the Dataset:

Step 1: Click on "Microsoft Excel" or "Text File"

On the Start Page, under Connect, you will see options like:

Microsoft Excel (for .xlsx)

Text File (for .csv)

Click the correct option.

Step 2: Select Your File

Browse your computer.

Select your dataset file.

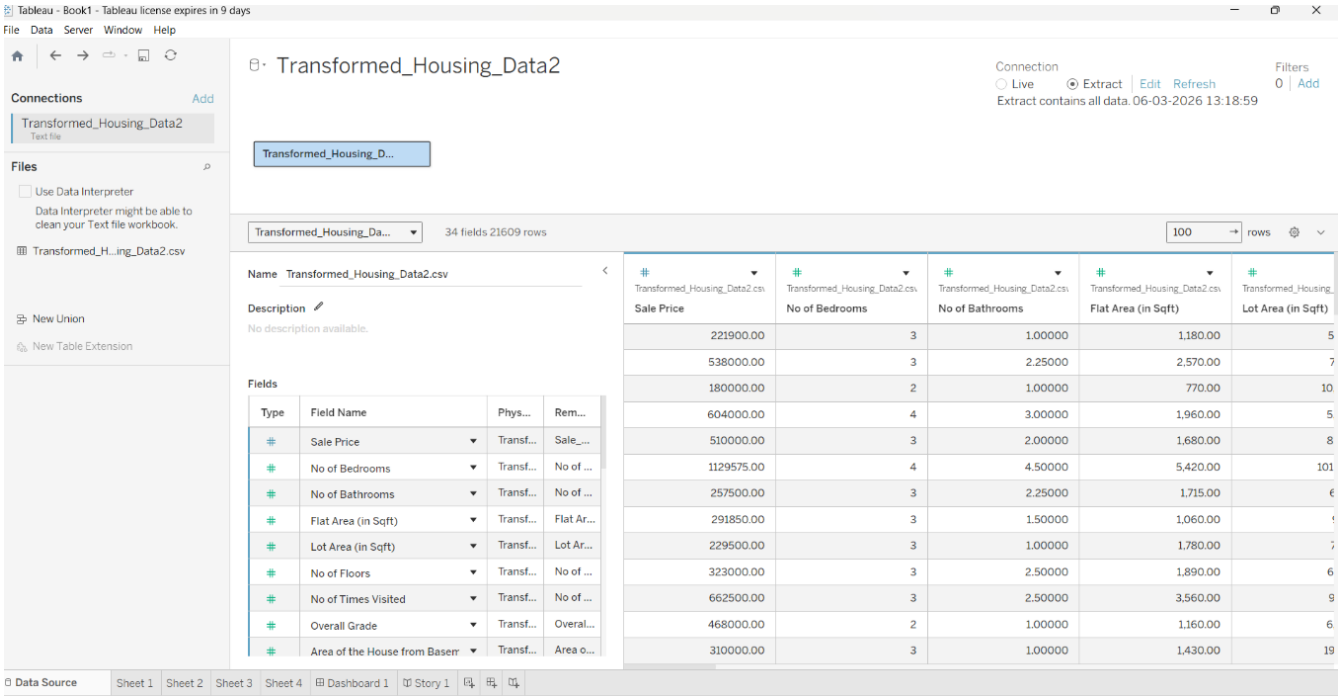
Click Open.

Step 3: Data Source Page

Your sheet names will appear on the left side.

Drag the sheet to the centre area.

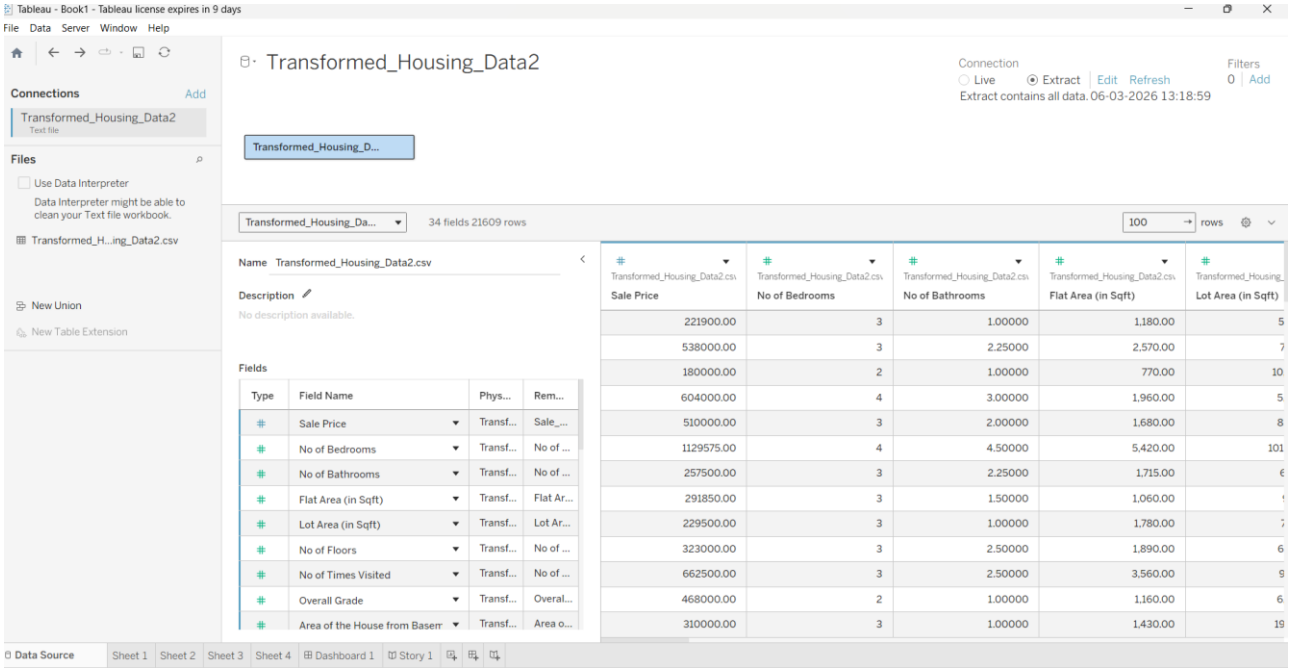
Tableau will show a preview of your data.



2) Data Preparation:

❖ Data Cleaning

Preparing the data for visualization involves cleaning the data to remove irrelevant or missing data, transforming the data into a format that can be easily visualized, exploring the data to identify patterns and trends, filtering the data to focus on specific subsets of data, preparing the data for visualization software, and ensuring the data is accurate and complete.



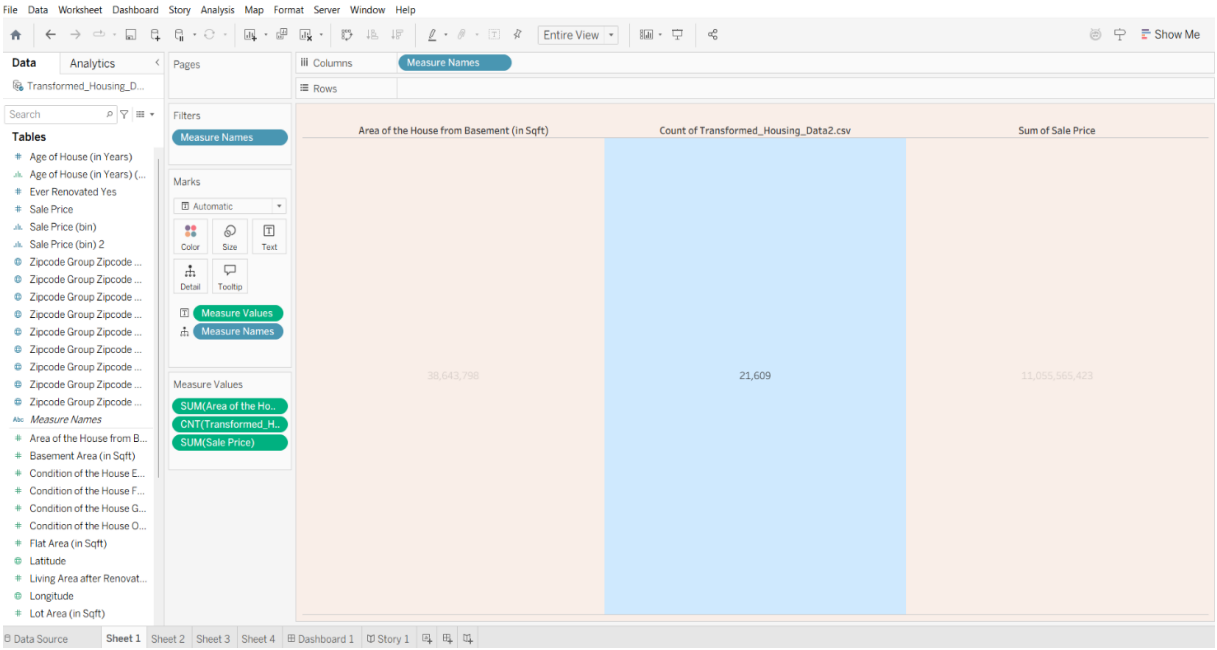
3) Data Visualization:

Data visualization is the process of creating graphical representations of data to help people understand and explore the information. The goal of data visualization is to make complex data sets more accessible, intuitive, and easier to interpret. By using visual elements such as charts, graphs, and maps, data visualizations can help people quickly identify patterns, trends, and outliers in the data.

❖ No of Unique Visualizations:

1) Count of Transformed_Housing_Data2.cvs

a) Drag and Drop the Measure Name and Measure Values.
b) Add only-
   - Count of Transformed_Housing_Data2.cvs
   - Area of the House from Basement (in Sqft)
   - Avg.SalePrice
c) Drag and Drop Measure Name at Column section.



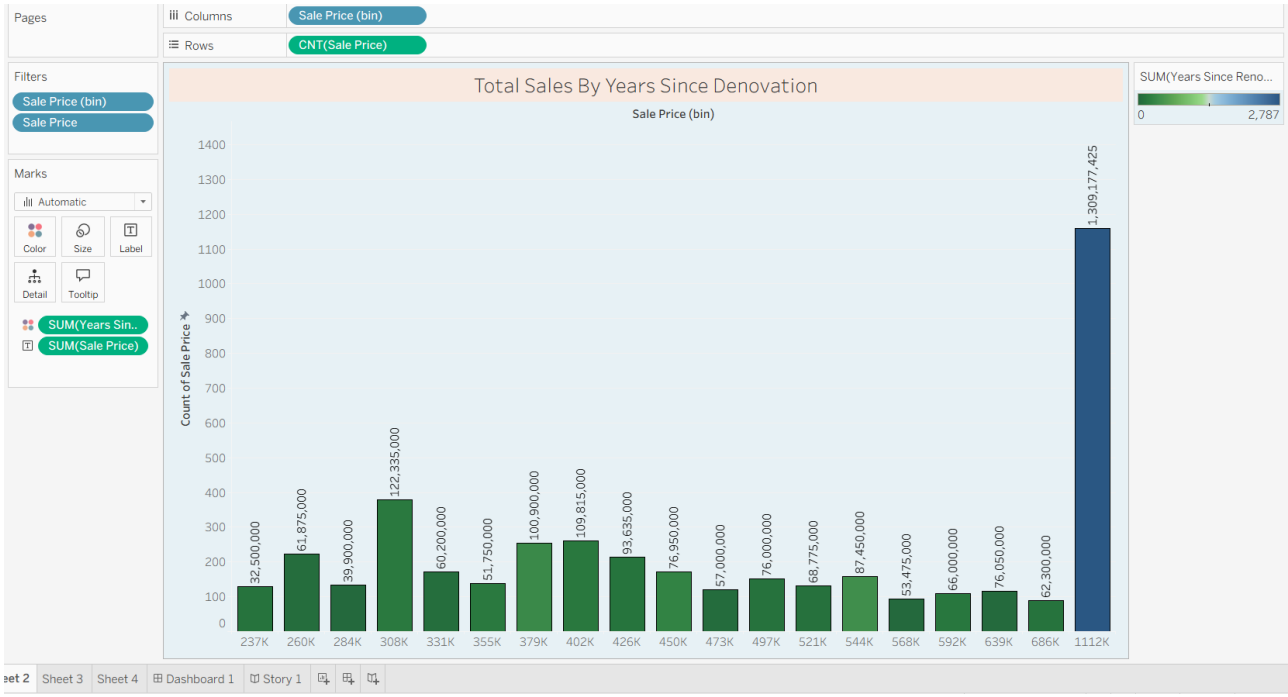
2) Total Sales by Years Since Renovation

a) Convert the Sales Price into Dimension by right clicking.

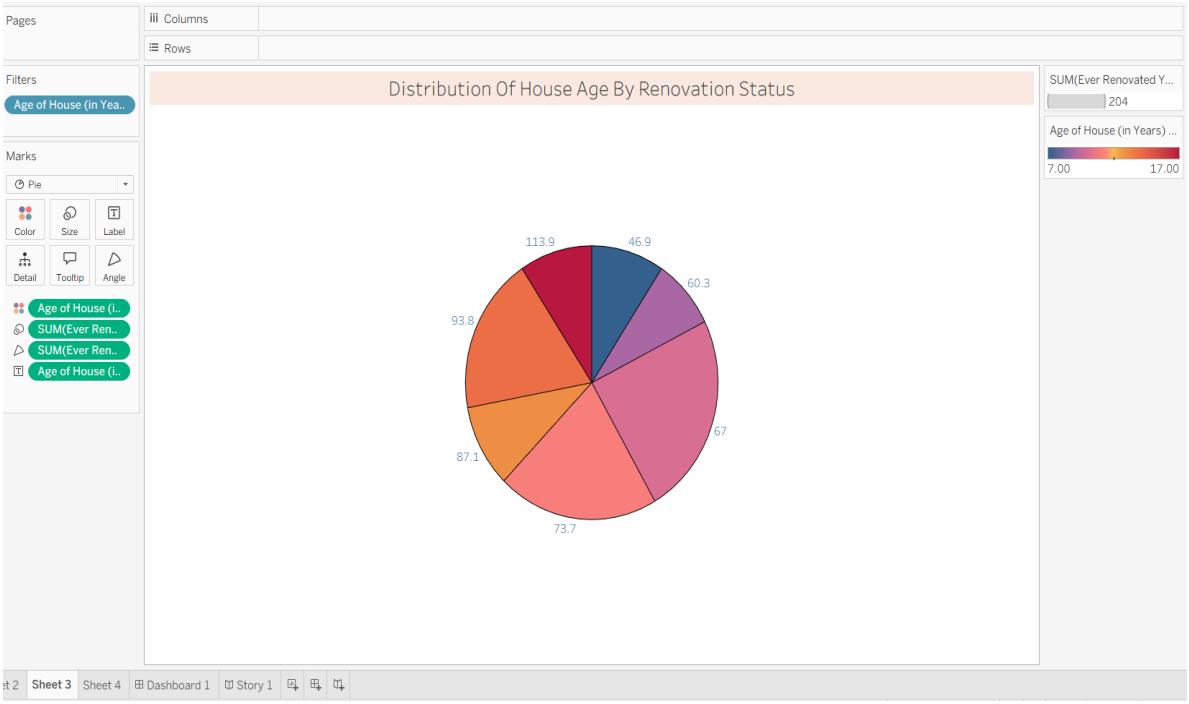
b) Add Filters into Sales Price and Sales Price(bin) to display Top 25 records.

c) Drag and Drop Sales Price(bin) for grouping Sales Price into column section, and also CNT(Sales price) into rows.

d) Drag and Drop SUM(Year Since Renovation) into color and SUM(Sale Price) into Lable.



3) **Distribution of House Age by Renovation Status**
   a) Convert Age of House(in Years) to Dimension.
   b) Filter Age of House(in Years) into Top 10 records.
   c) Add Age of House(in Years) into color.
   d) Add SUM(Ever Renovated yes) into angle.
   e) Add Sum(Ever Renovated yes) into size.
   f) Add Age of House(in Years) in to label.



4) House Age Distribution by Number of Bathrooms, Bedrooms and Floors

a) Drag and Drop Measure Name and Age of House(in Years).

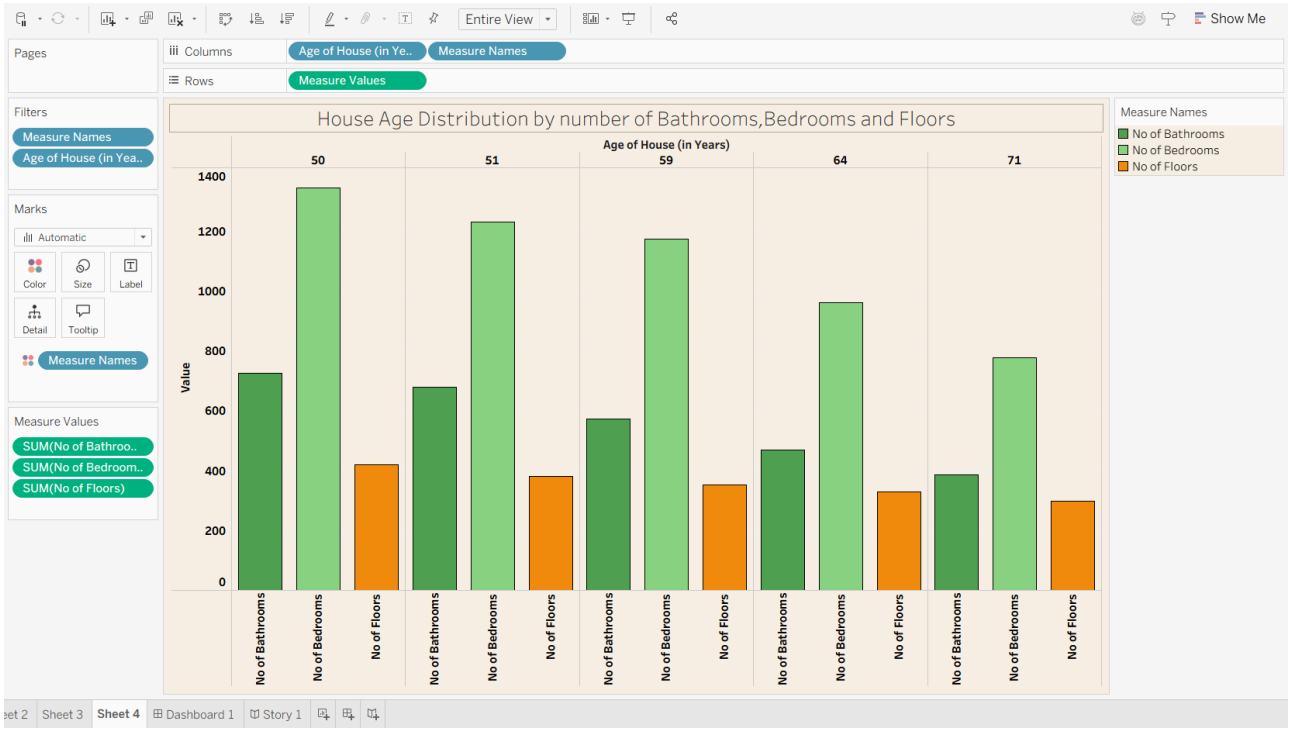
b) Filter Age of House(in Years) into Top 5 records.

c) Drag and Drop Measure Values-

- SUM(No of Bathroom)
- SUM(No of Bedroom)
- SUM(No of Floors)

d) Drag and Drop Measure Names into Color.

e) Add Age of House(in Year) and Measure Name into column section, and Measure Values into rows.

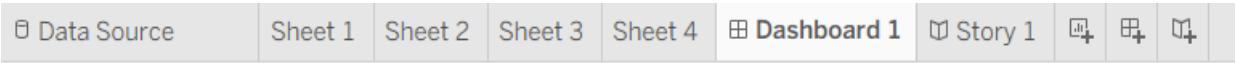


4) Dashboard

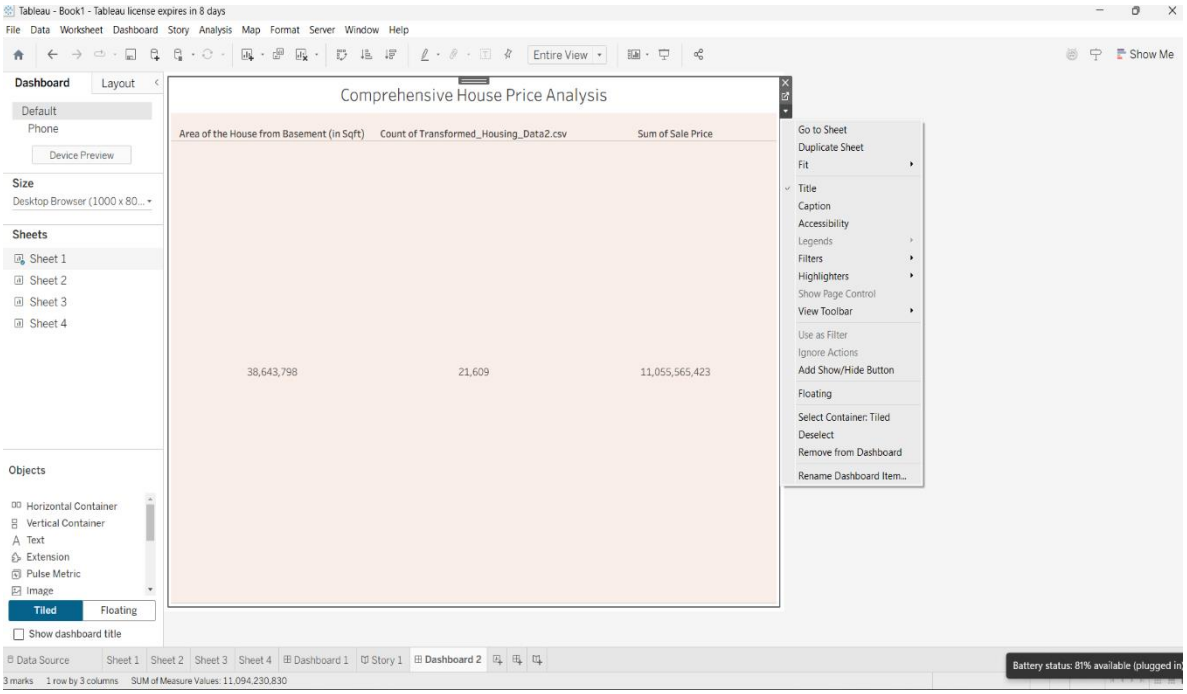
A dashboard is a graphical user interface (GUI) that displays information and data in an organized, easy-to-read format. Dashboards are often used to provide real-time monitoring and analysis of data and are typically designed for a specific purpose or use case. Dashboards can be used in a variety of settings, such as business, finance, manufacturing, healthcare, and many other industries. They can be used to track key performance indicators (KPIs), monitor performance metrics, and display data in the form of charts, graphs, and tables.

❖ Responsive and Design of Dashboard

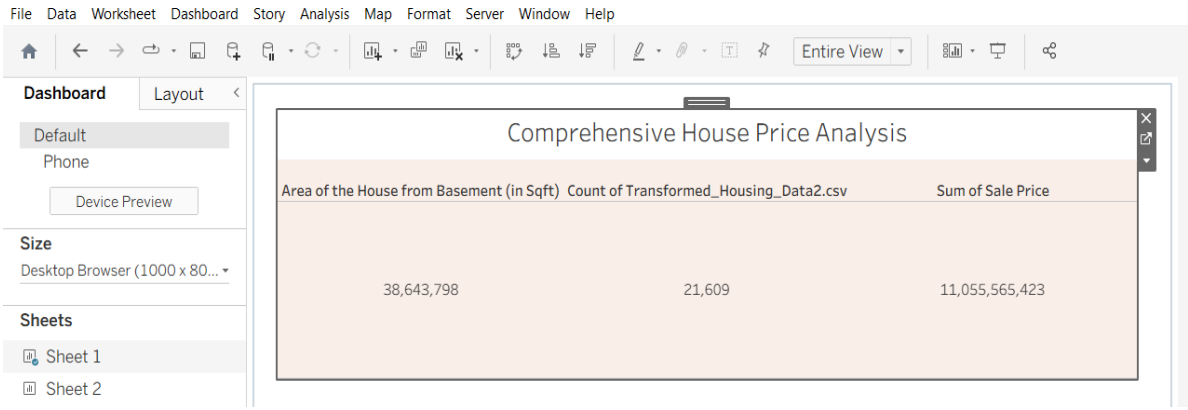
a) Click On **New Dashboard** option.



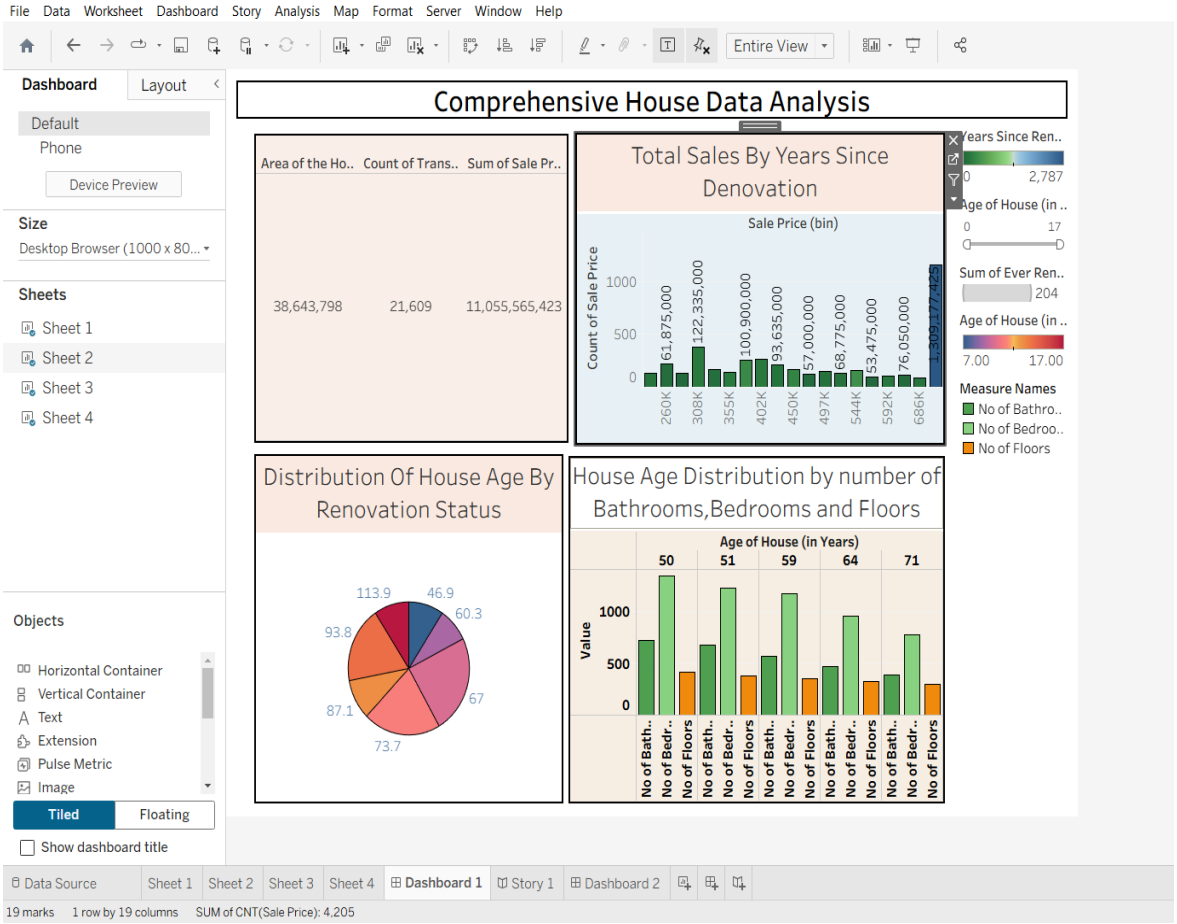
b) Drag and Drop Sheets from the data panel into centre white screen and in the right side of sheet there is Downward arrow so click on that and choose **Floating** option.



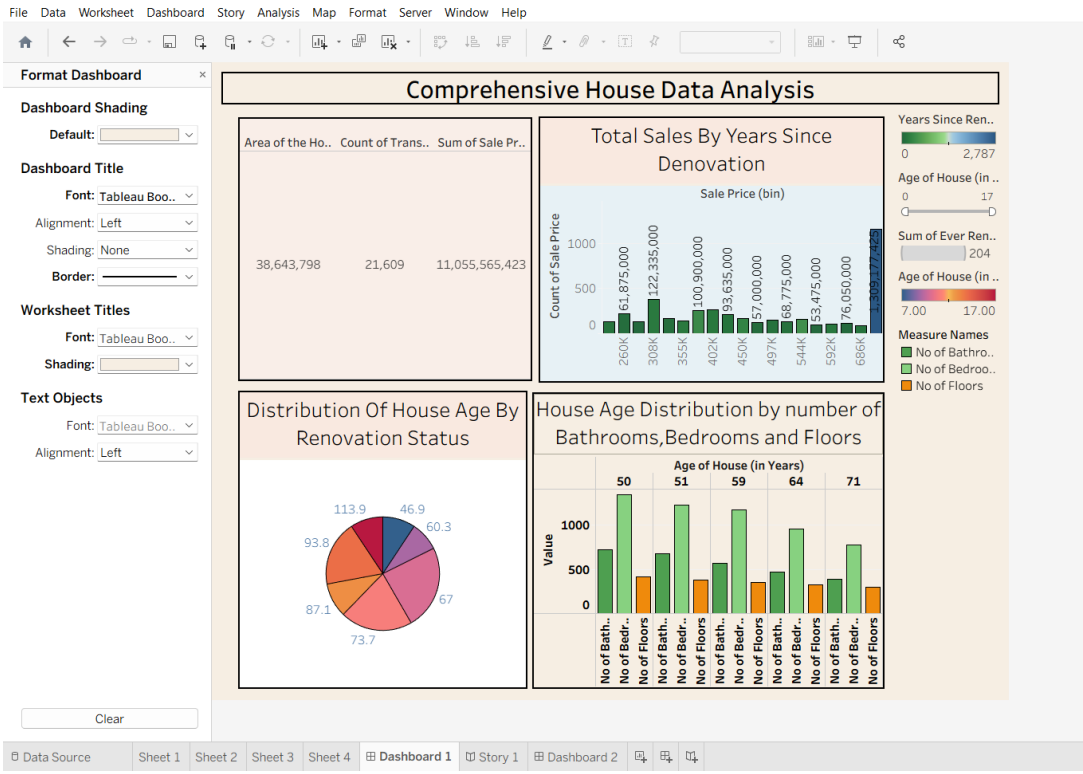
c) After selecting Floating option it's looks like.



d) Drag and Drop the Required Sheets for dashboard there I added 4 sheets.



e) Add a Format to an Dashboard by clicking at Format option at the Top and add background color to dashboard and boarder to the sheets.

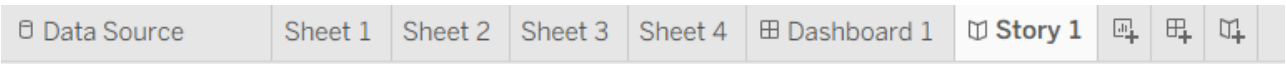


5) Storyboard

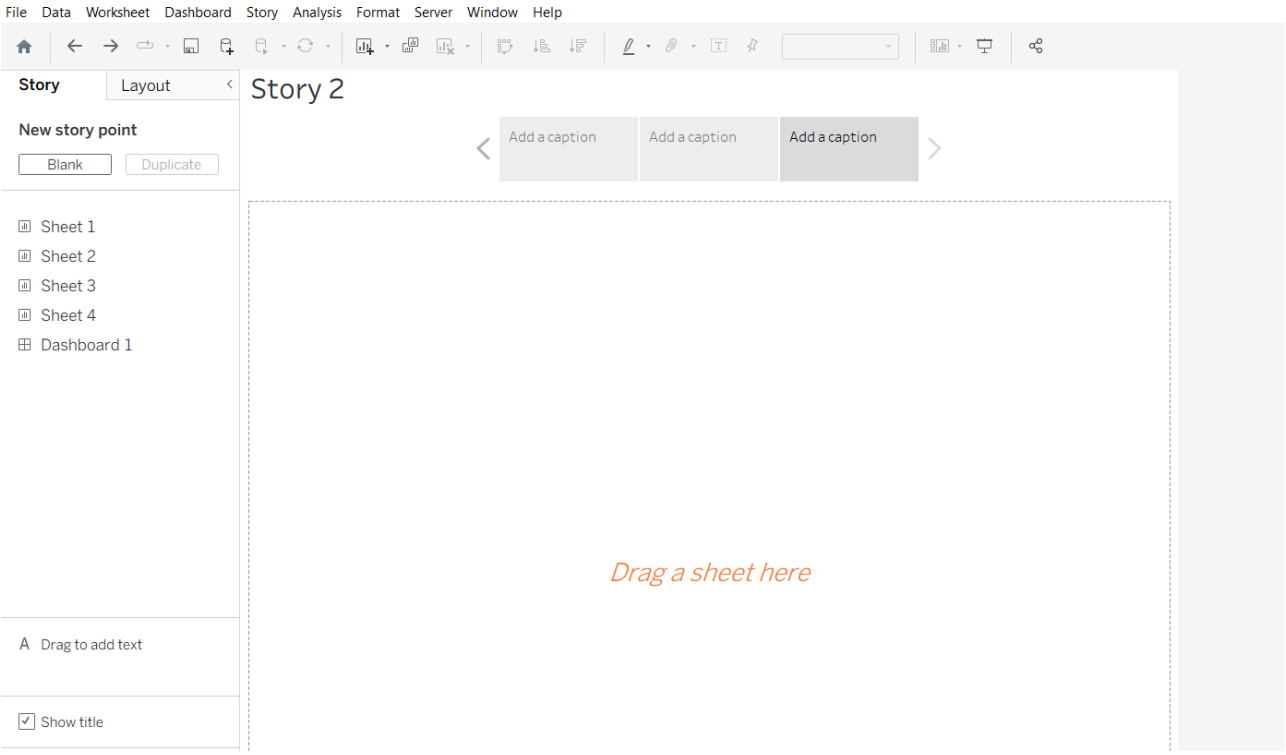
A storyboard is a visual representation of a sequence of events, typically used in multimedia projects such as films, animations, advertisements, or presentations. It consists of a series of drawings or images arranged in a sequence, often accompanied by annotations or descriptions, to outline the flow of the story or concept.

❖ Design a Storyboard

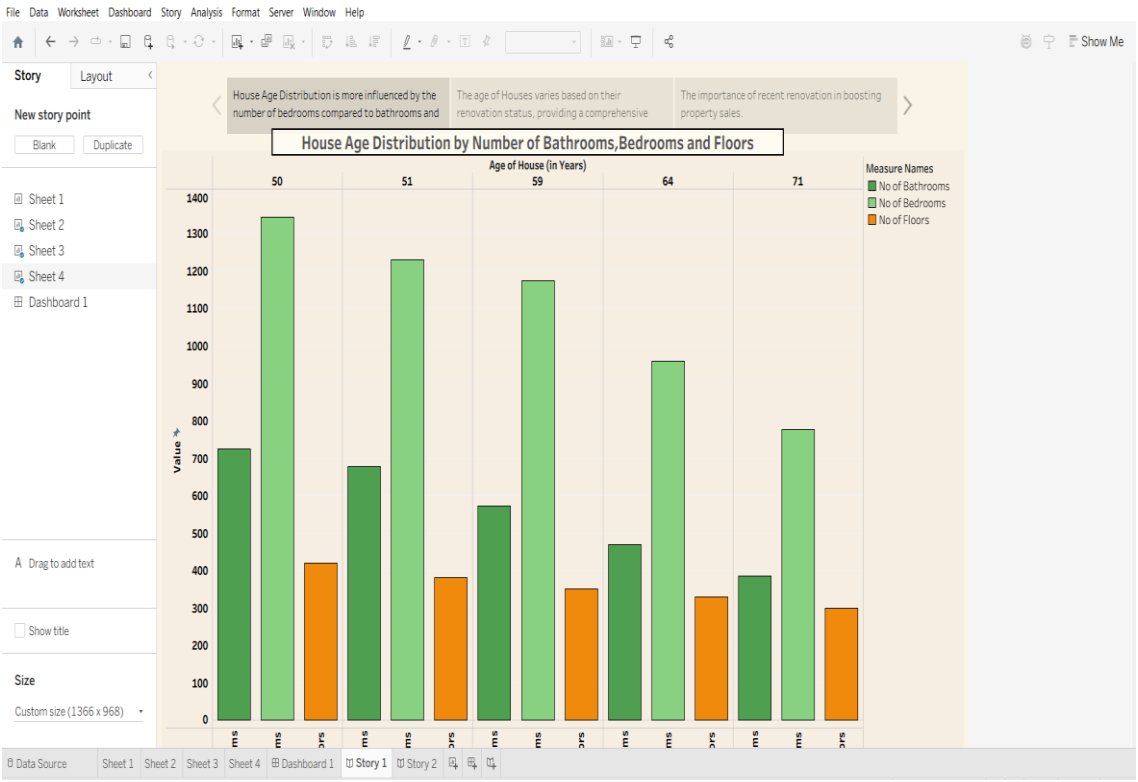
a) Click On **New Story** option.

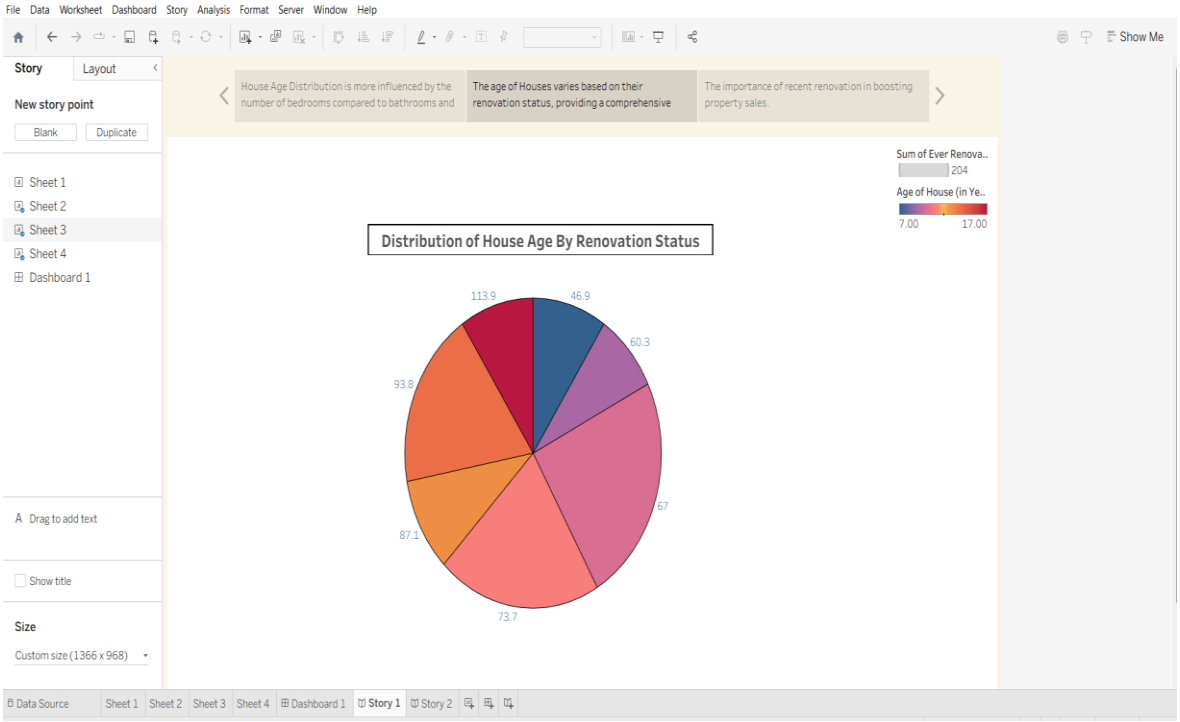


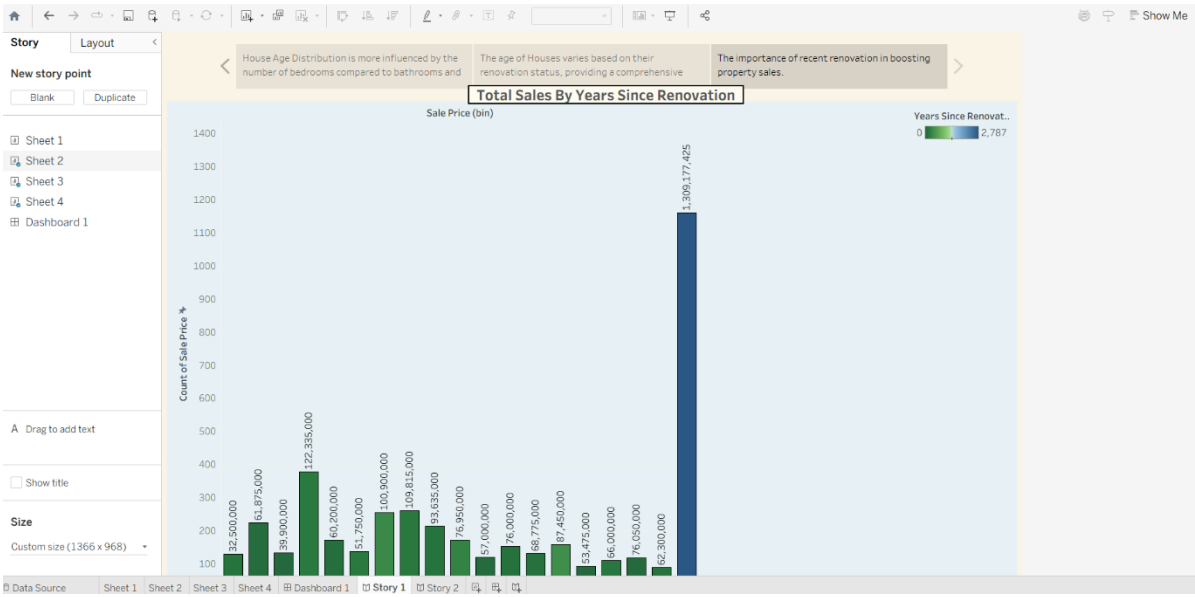
b) Here I have to add 3 sheets into one story so click on **Blank** .



c) Drag and Drop Sheets from Story panel into white Screen and add caption for each sheet.




File Data Worksheet Dashboard Story Analysis Format Server Window Help
Show Me
Story
Layout
New story point
Blank
Duplicate
Sheet 1
Sheet 2
Sheet 3
Sheet 4
Dashboard 1
Drag to add text
Show title
Size
Custom size (1366 x 968)
House Age Distribution is more influenced by the number of bedrooms compared to bathrooms and
The age of Houses varies based on their renovation status, providing a comprehensive
The importance of recent renovation in boosting property sales.
Sum of Ever Renova..
204
Age of House (in Ye..
7.00
17.00
Distribution of House Age By Renovation Status
46.9
60.3
67
73.7
87.1
93.8
113.9
Data Source
Sheet 1
Sheet 2
Sheet 3
Sheet 4
Dashboard 1
Story 1
Story 2


Show Me
Story
Layout
New story point
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Duplicate
Sheet 1
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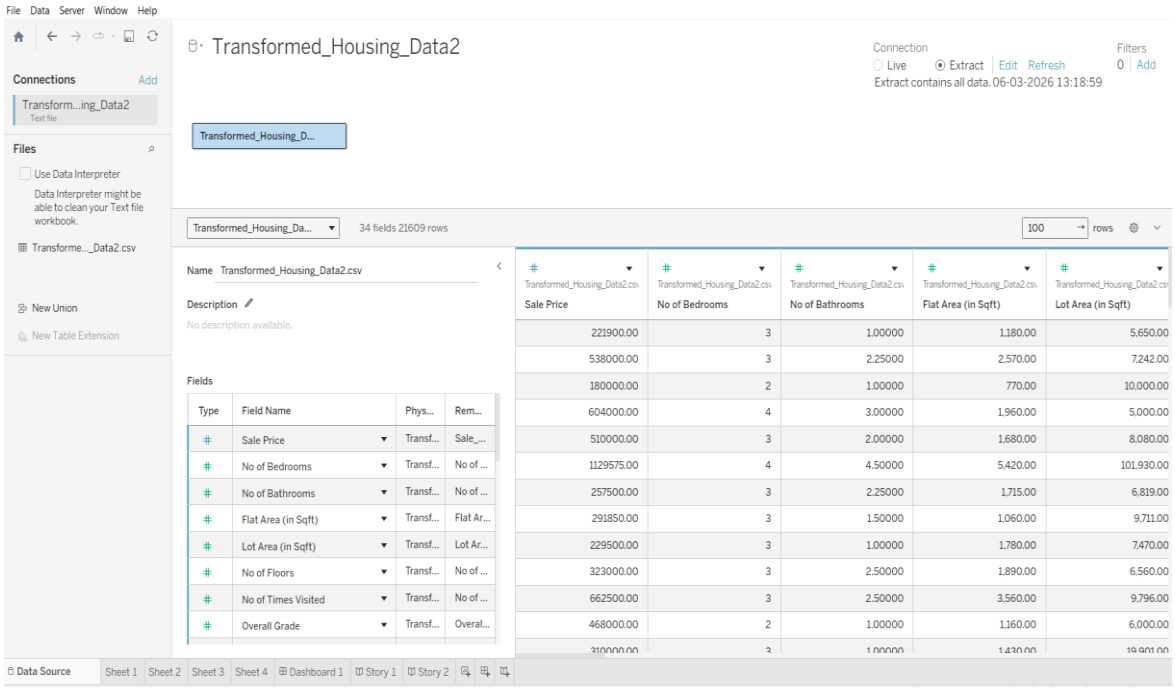
Dashboard 1
Drag to add text
Show title
Size
Custom size (1366 x 968)
House Age Distribution is more influenced by the number of bedrooms compared to bathrooms and
The age of Houses varies based on their renovation status, providing a comprehensive
The importance of recent renovation in boosting property sales.
Total Sales By Years Since Renovation
Sale Price (bin)
Years Since Renovat..
0
2,787
Count of Sale Price
1400
1300
1200
1100
1000
900
800
700
600
500
400
300
200
100
32,500,000
61,875,000
39,900,000
122,335,000
60,200,000
51,750,000
100,900,000
109,815,000
93,635,000
76,950,000
57,000,000
76,000,000
68,775,000
87,450,000
53,475,000
66,000,000
76,050,000
62,300,000
1,309,177,425
Data Source
Sheet 1
Sheet 2
Sheet 3
Sheet 4
Dashboard 1
Story 1
Story 2

6) Performance Testing

Performance testing involves assessing the volume of data rendered from the database, the impact of data filters on system responsiveness, and the complexity introduced by the number of visualizations. Optimizing these factors ensures the dashboard operates efficiently, providing timely and reliable insights.

❖ Amount of Data Loaded

"Amount of Data Loaded" refers to the quantity or volume of data that has been imported, retrieved, or loaded into a system, software application, database, or any other data storage or processing environment. It's a measure of how much data has been successfully processed and made available for analysis, manipulation, or use within the system.

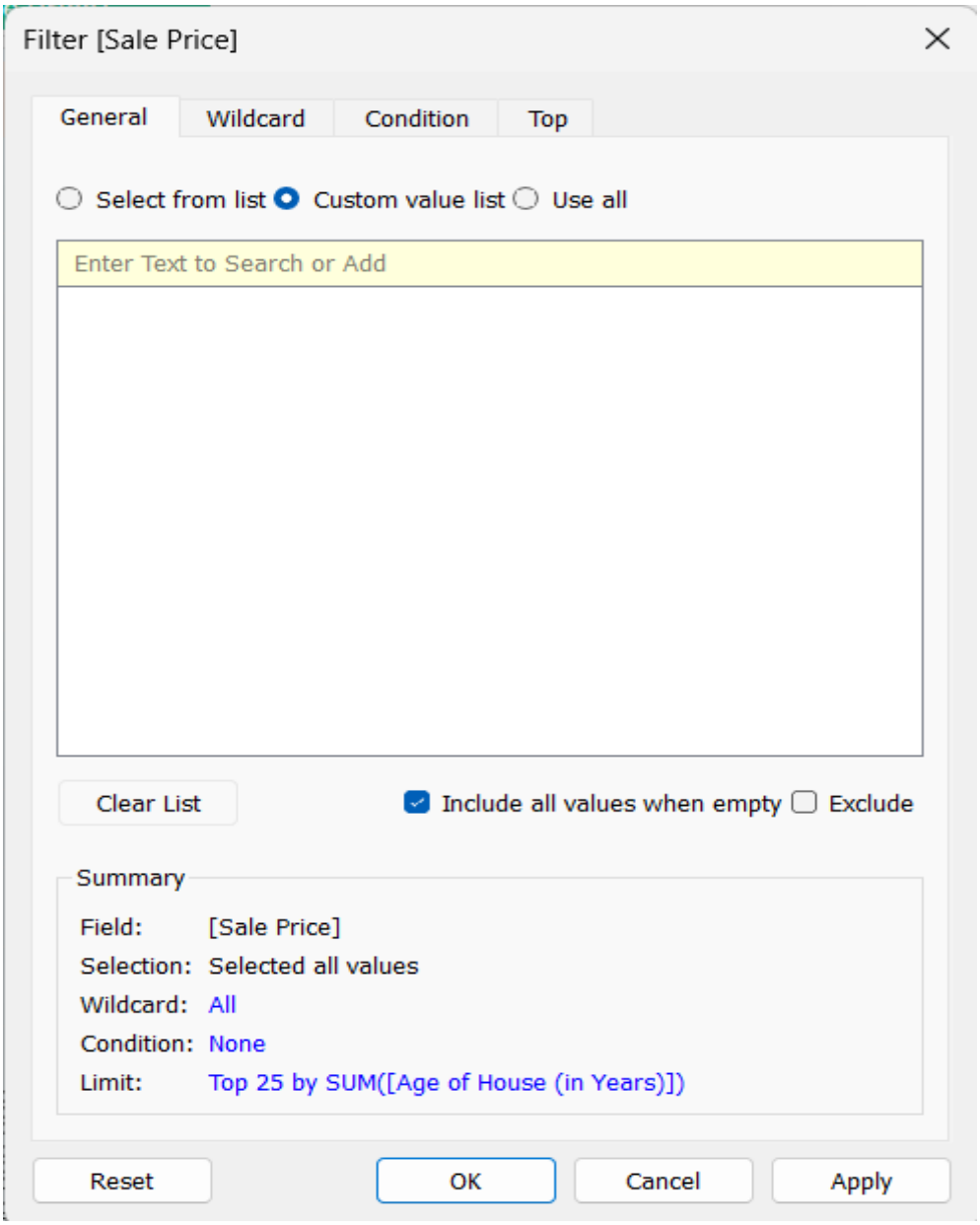


❖ Utilization of Filters

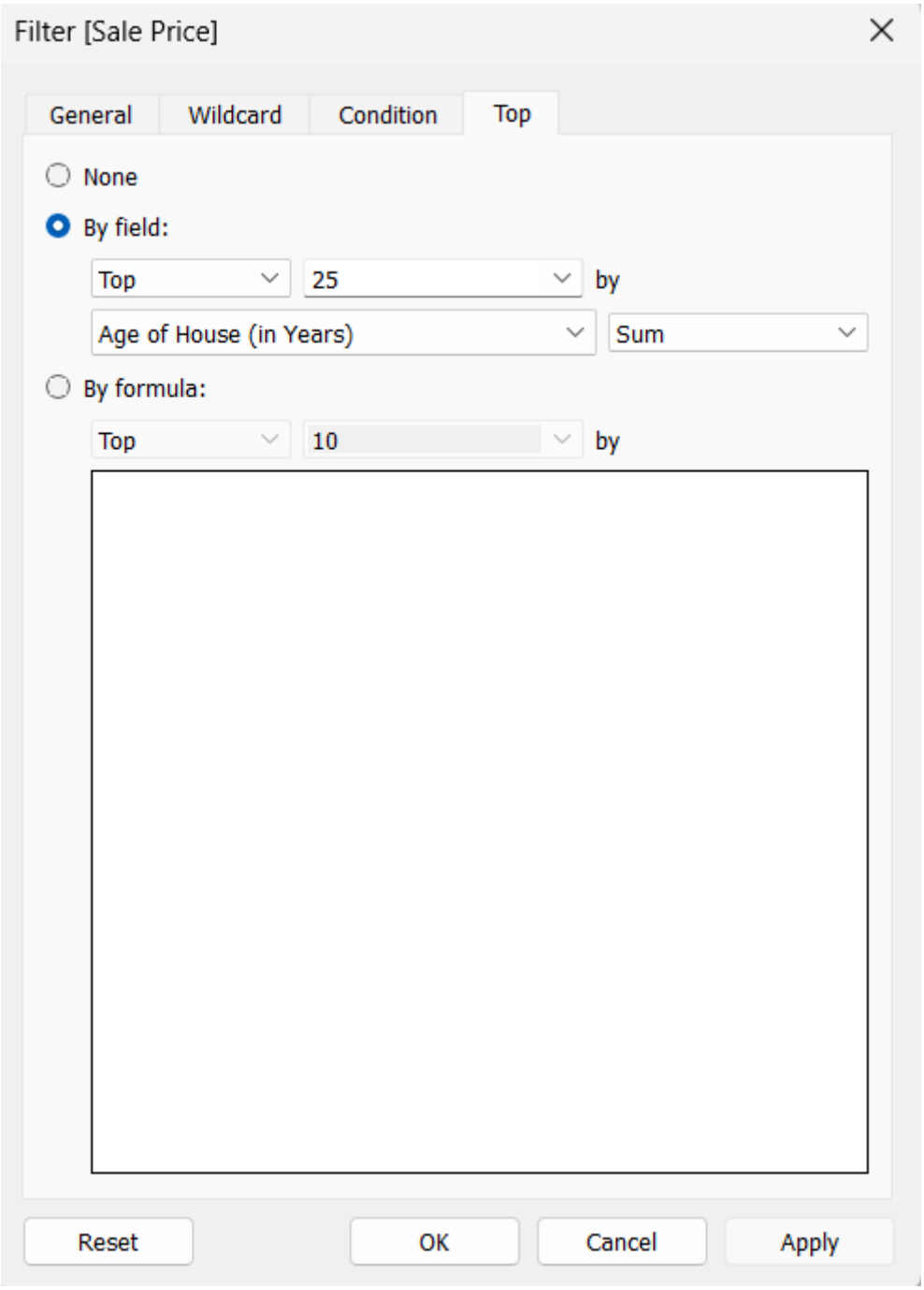
"Utilization of Filters" refers to the application or use of filters within a system, software application, or data processing pipeline to selectively extract, manipulate, or analyse data based on specified criteria or conditions.

1) Selected "Sales Price" as a Top N Filters.

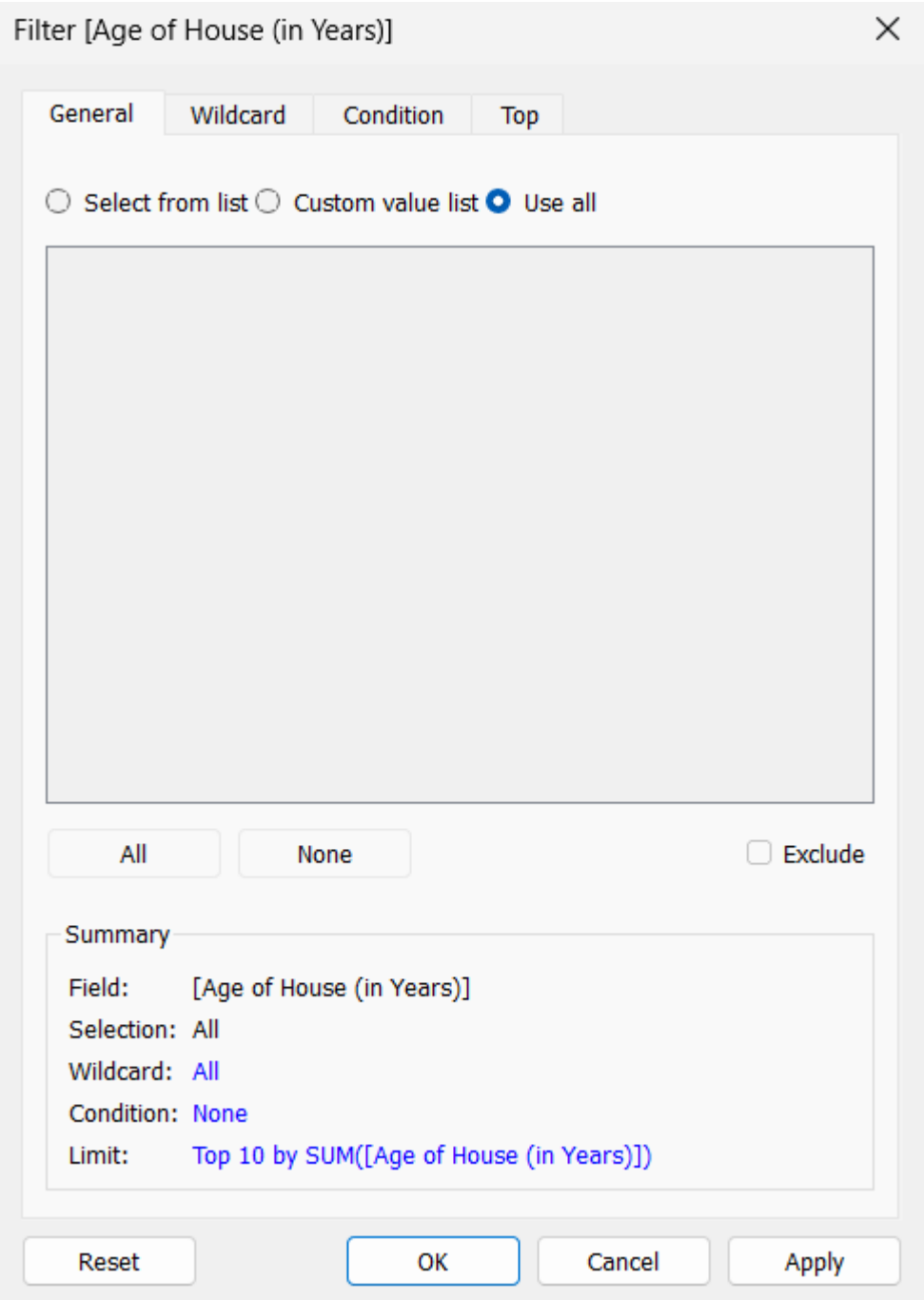
a) Drage and drop "Sales Price" into Filters Filed.
b) Right click on "Sales Price" and click on **Edit Filter**.
c) Then Click on **Top/Bottom**.



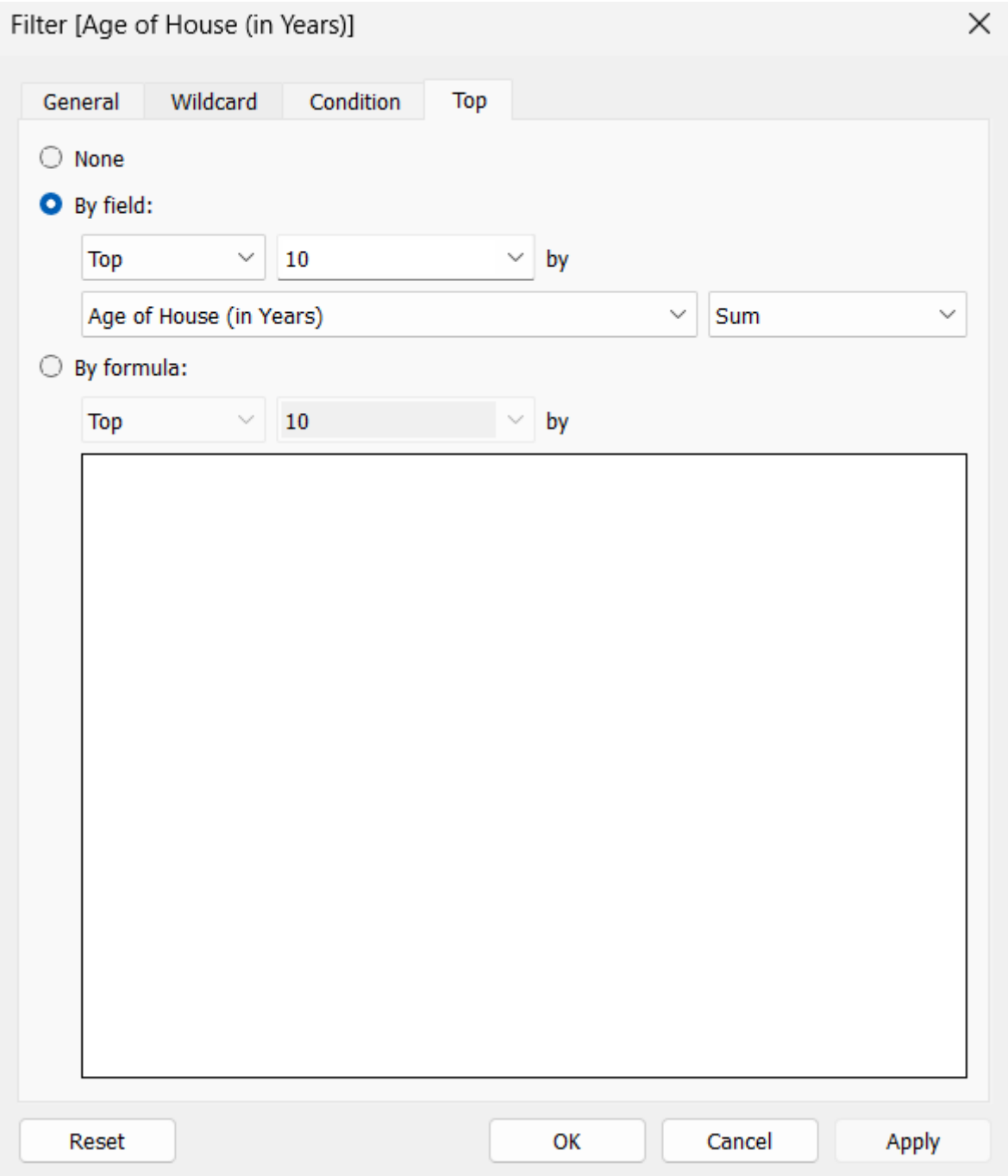
d) Select **By Field** and add Top 25 records and click OK.



2) **Select "Age of House (in Year)" as a Top N records.**
   a) Drage and drop "Age of House (in Year)" into Filters Filed.
   b) Right click on "Sales Price" and click on **Edit Filter**.
   c) Then Click on **Top/Bottom**.

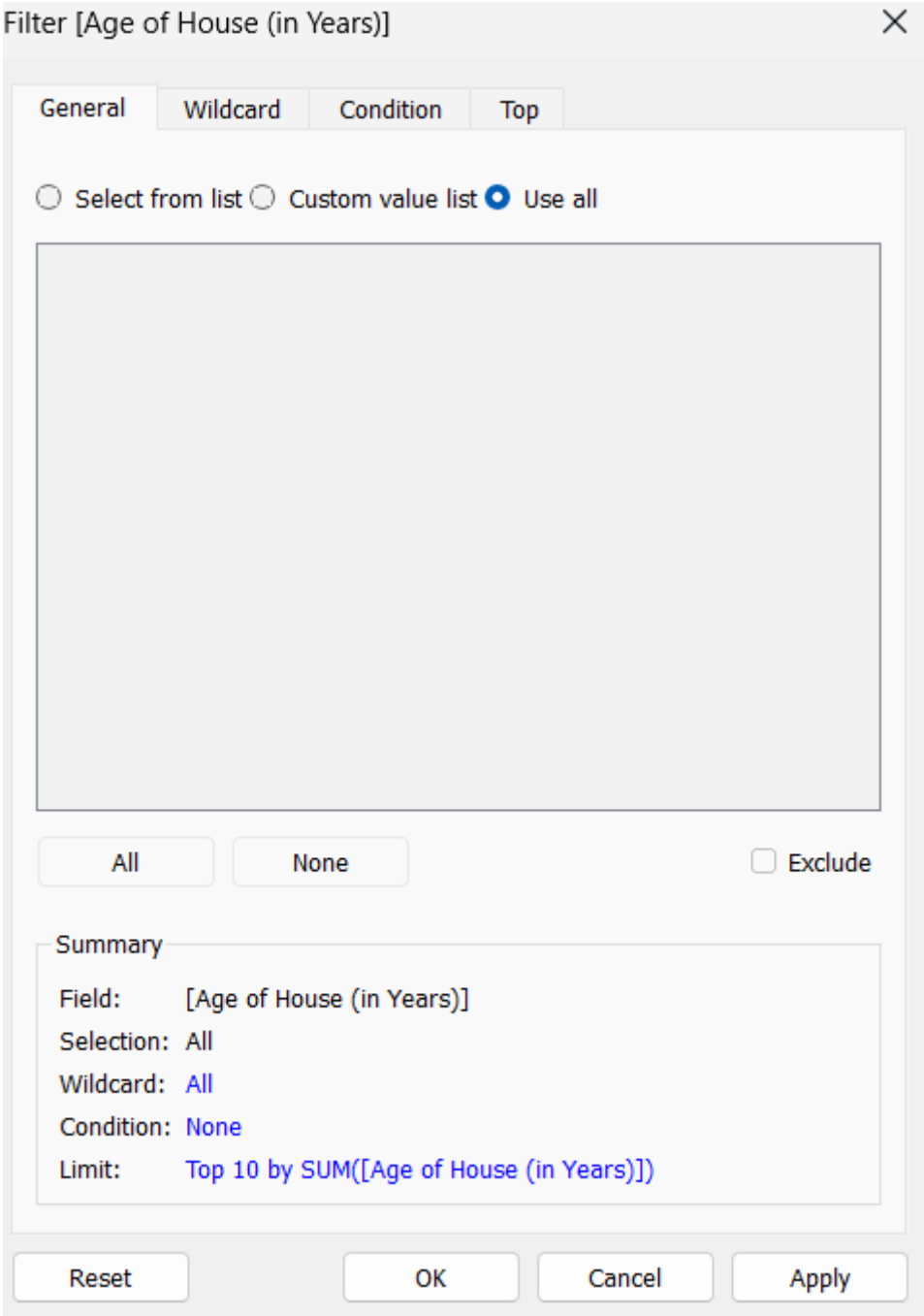


d) Select **By Field** and add Top 10 records and click OK.

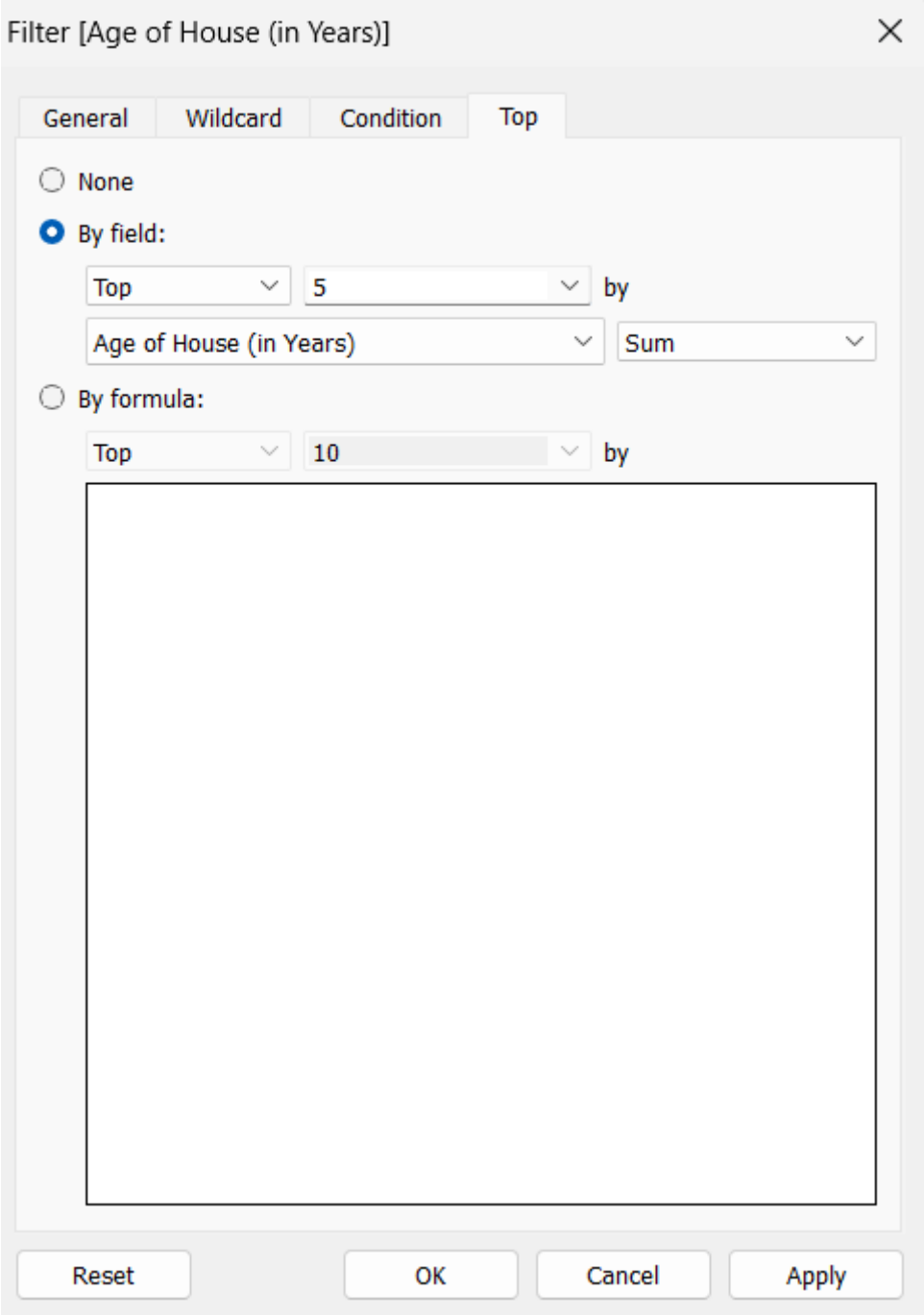


3) **House Age Distribution by Top N Filters.**

a) Drage and drop “Age of House (in Year)” into Filters Filed.
b) Right click on “Sales Price” and click on **Edit Filter**.
c) Then Click on **Top/Bottom**.

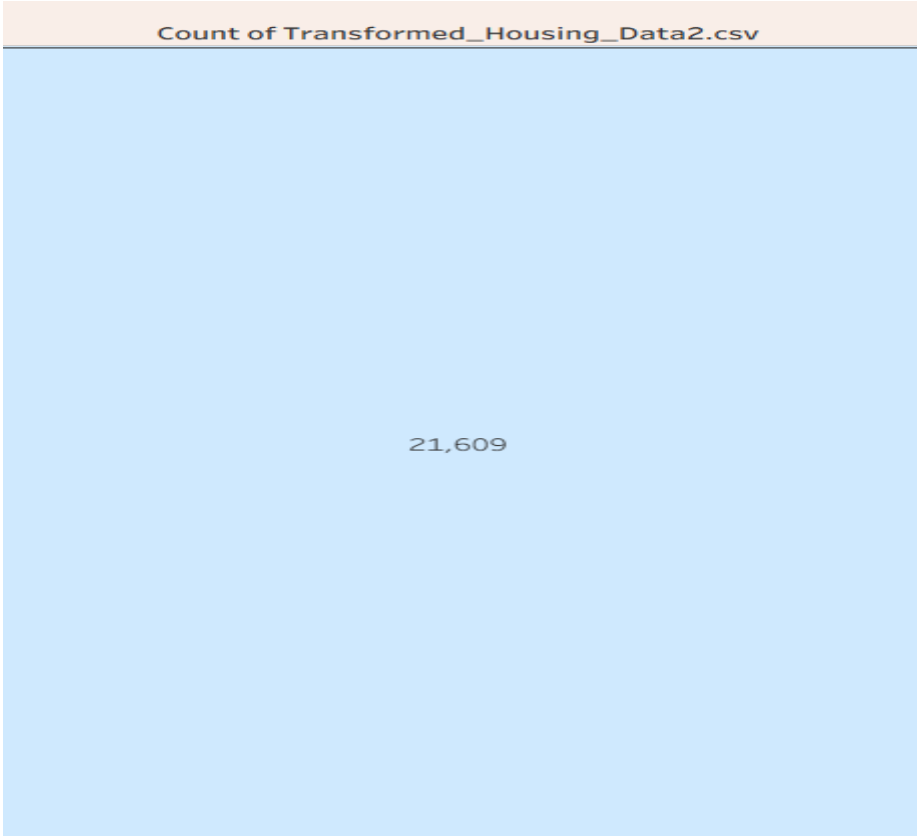


d) Select **By Field** and add Top 5 records and click OK.



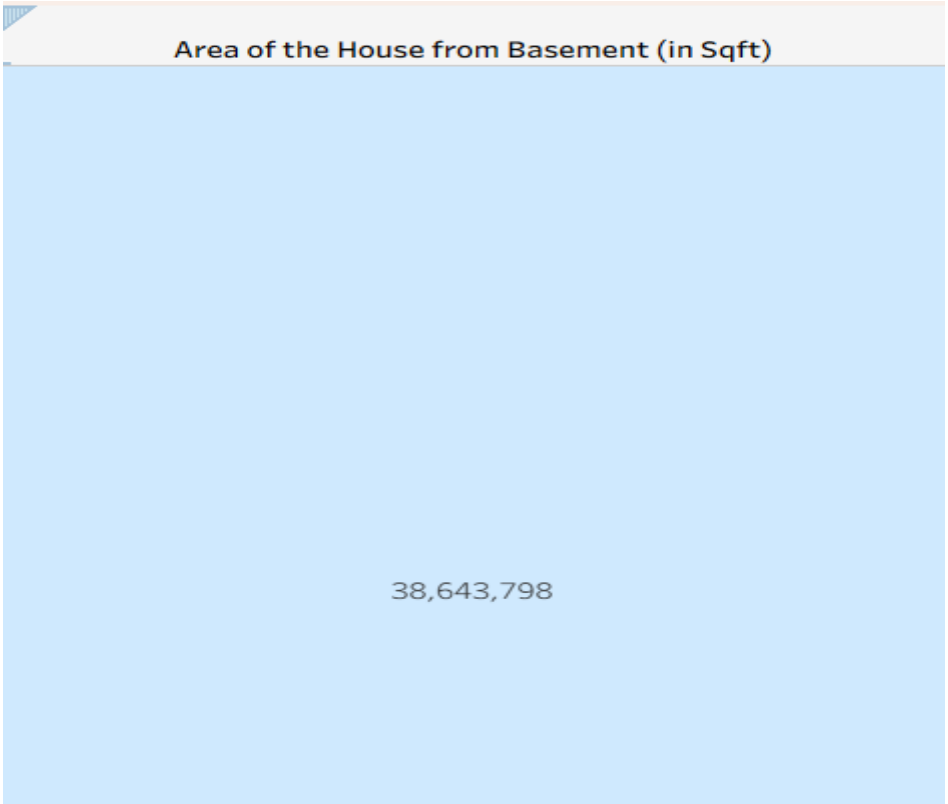
❖ No of Visualizations/ Graphs

1) Count of Transformed Housing Data

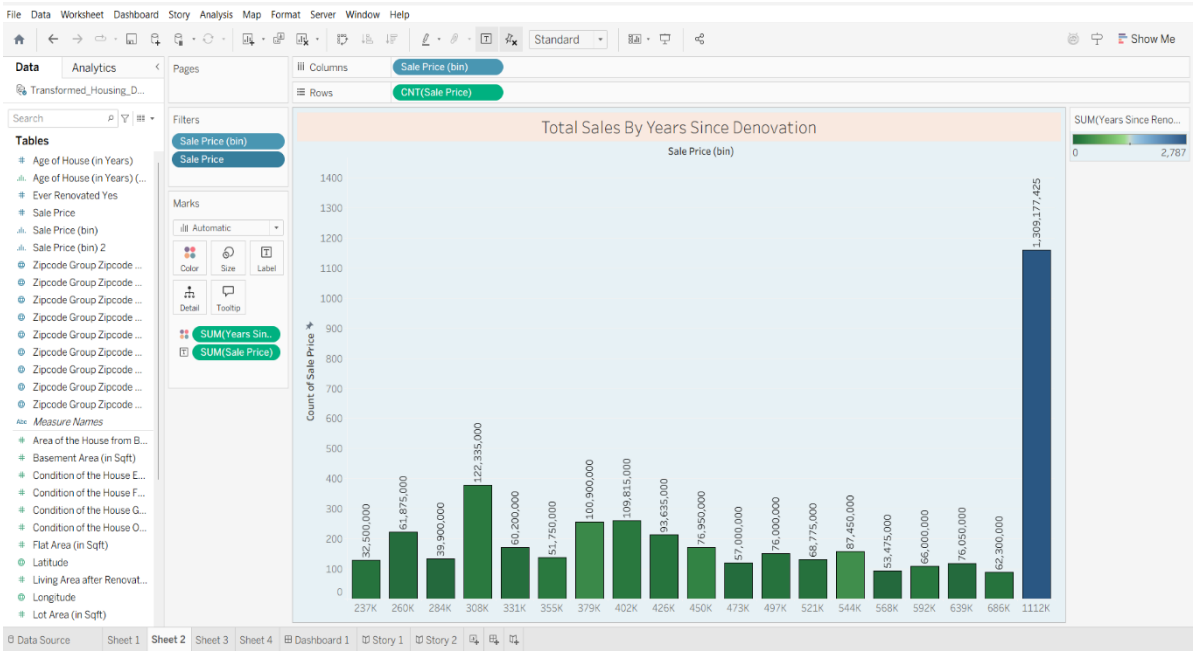


2) Average Sale Prices

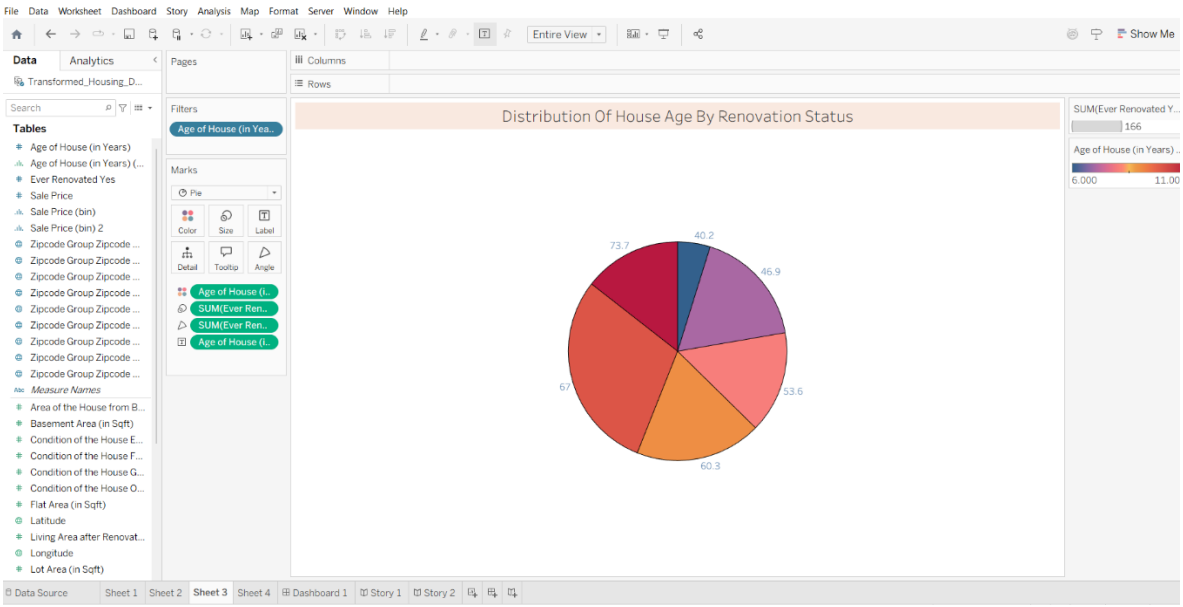
3) Area of House from Basement (in Sqft)



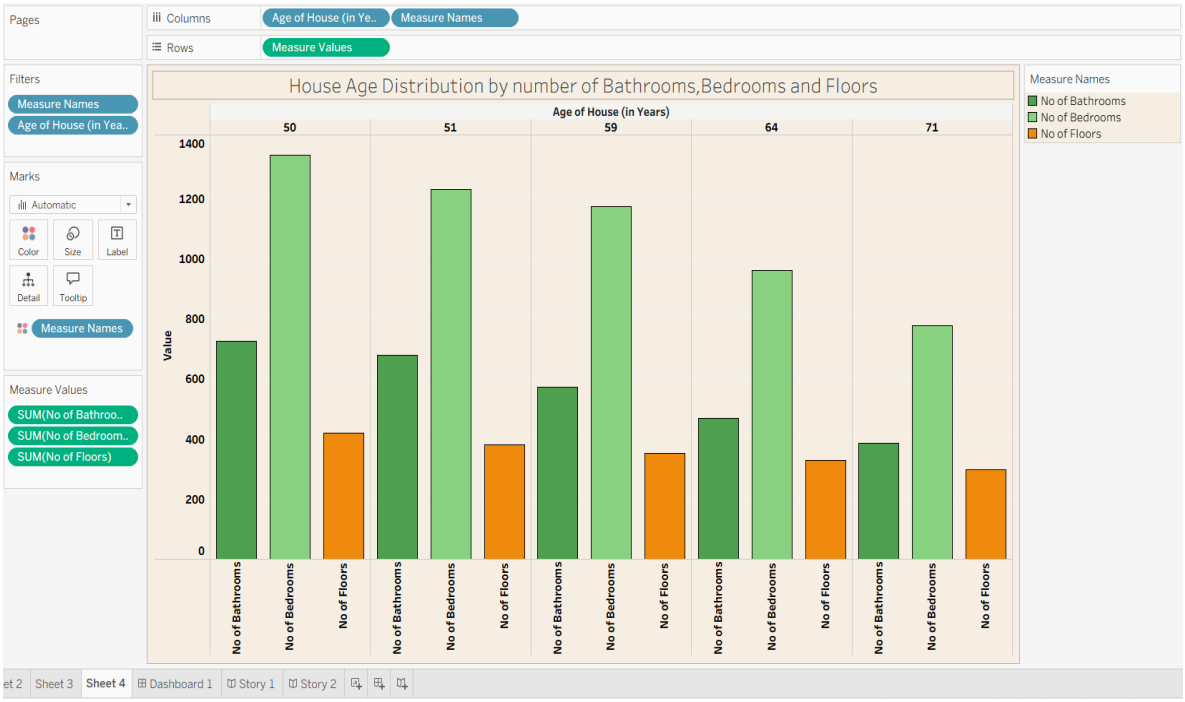
4) Total Sales by Years Since Renovation



5) Distribution of House Age by Renovation Status



6) House Age Distribution by Number of Bathrooms, Bedrooms, and Floors



7) Web integration

Web integration of a Tableau involves embedding interactive visualizations into a website or web application. This allows users to explore data insights directly within a web interface, enhancing accessibility and engagement. It supports real-time updates, user filtering, and seamless navigation for a dynamic data storytelling experience.

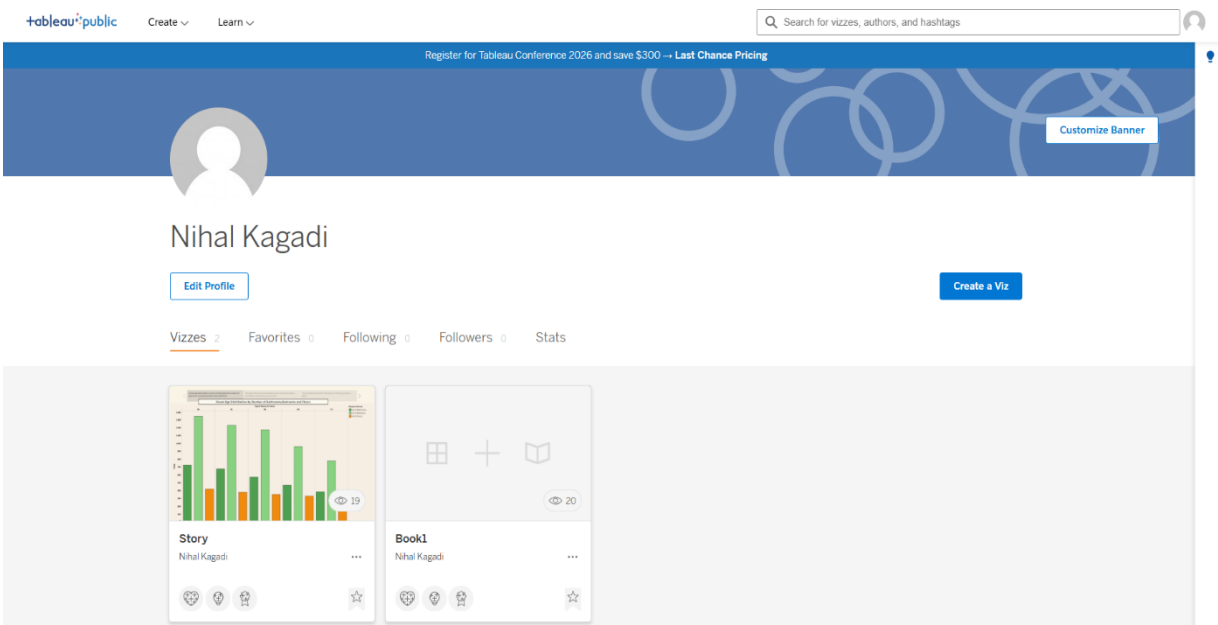
❖ Publishing

a) **Sign in to Tableau Public**

In Tableau Desktop, go to **File** > **Save to Tableau Public**.
If you're not already signed in, a login window will appear.
Enter your **Tableau Public credentials** or sign up if you don't have an account.

b) **Save and Publish**

- After logging in, you'll be prompted to **name your workbook**.
- Click **Save** – Tableau will upload the workbook to your Tableau Public profile.



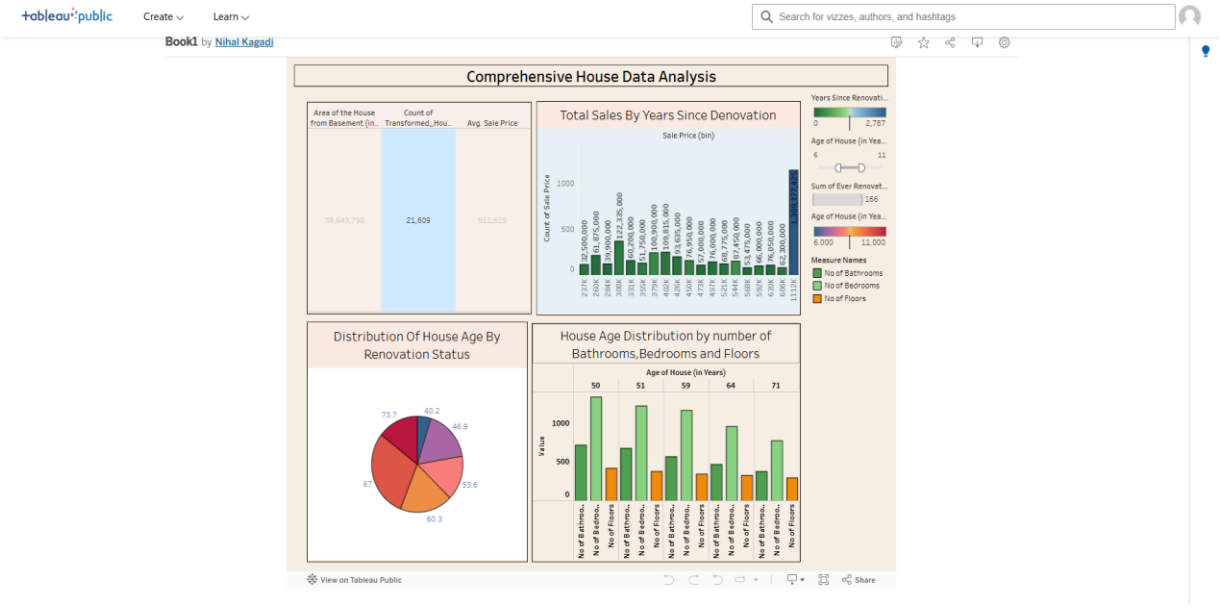


tableau public
Create
Learn
Search for vizzes, authors, and hashtags
Book1 by Nihal Kagadi
Comprehensive House Data Analysis
Area of the House from Basement (in...
Count of Transformed_Hou...
Avg. Sale Price
38,643,798
21,609
511,619
Total Sales By Years Since Denovation
Sale Price (bin)
Count of Sale Price
Years Since Renovati...
Age of House (in Yea...
Sum of Ever Renovat...
Measure Names
No of Bathrooms
No of Bedrooms
No of Floors
Distribution Of House Age By Renovation Status
House Age Distribution by number of Bathrooms,Bedrooms and Floors
Age of House (in Years)
Value
View on Tableau Public
Share

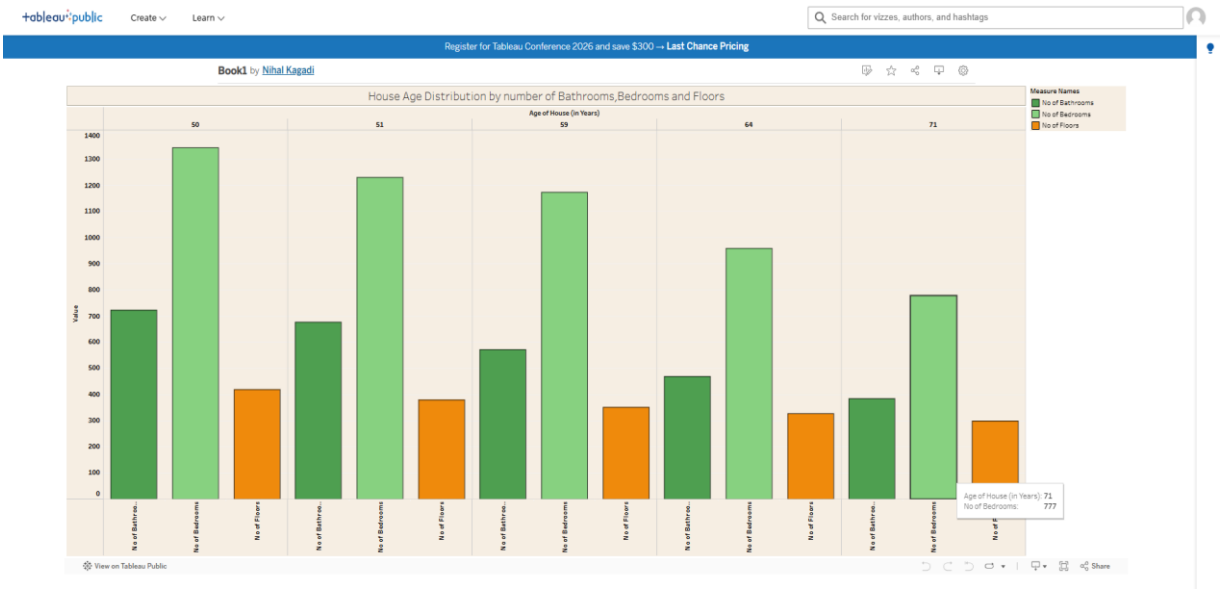


tableau public
Create
Learn
Search for vizzes, authors, and hashtags
Register for Tableau Conference 2026 and save $300 → Last Chance Pricing
Book1 by Nihal Kagadi
House Age Distribution by number of Bathrooms,Bedrooms and Floors
Age of House (in Years)
Measure Names
No of Bathrooms
No of Bedrooms
No of Floors
Value
Age of House (in Years): 71
No of Bedrooms: 777
View on Tableau Public
Share

❖ Dashboard and Story embed with UI with Flask.

In this activity, you will create a simple web application using Flask and embed a Tableau Dashboard or Story into the UI. This allows users to view and interact with Tableau visualizations directly through a custom web interface.

1) temp.py

dashboard.html | app.py 1 | requirements.txt

app.py > ...

```python
from flask import Flask, render_template

app = Flask(__name__)

@app.route("/")
def home():
    return render_template("index.html")

@app.route("/dashboard")
def dashboard():
    return render_template("dashboard.html")

@app.route("/story")
def story():
    return render_template("story.html")

@app.route("/contact")
def contact():
    return render_template("contact.html")

if __name__ == "__main__":
    app.run(debug=True)
```

2) **Adding Dashboard and Story on HTML page.**

a) Open Tableau Public profile and open Dashboard/Story.

b) Click on share option and copy the embed code.

c) Paste in HTML page

app.py 1 | index.html | requirements.txt

index.html > ...

```html
<html>
<body>

<!-- Navigation Bar -->
<div class="navbar">
<a href="index.html">Home</a>
<a href="dashboard.html">Dashboard</a>
<a href="story.html">Story</a>
<a href="about.html">About</a>
</div>

<!-- Home Content -->
<div class="main">

<h1>Visualizing Housing Market Trends</h1>

<h3>An Analysis of Sale Prices and Features using Tableau</h3>

<p>
This project analyzes housing market data to understand trends in sale prices
and the impact of different property features. Interactive dashboards and
visualizations are created using Tableau to make the analysis easy to understand.
</p>
<!-- Home Content -->
<div class="hero">

<img src="https://images.unsplash.com/photo-1560518883-ce09059eeffa" alt="Housing Market" widt

<h1>Visualizing Housing Market Trends</h1>

<p>
An Analysis of Sale Prices and Features using Tableau.
This project explores housing market data using interactive dashboards
to understand trends in property prices and housing features.
</p>

</div>
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Home Dashboard Story About
Visualizing Housing Market Trends
An Analysis of Sale Prices and Features using Tableau
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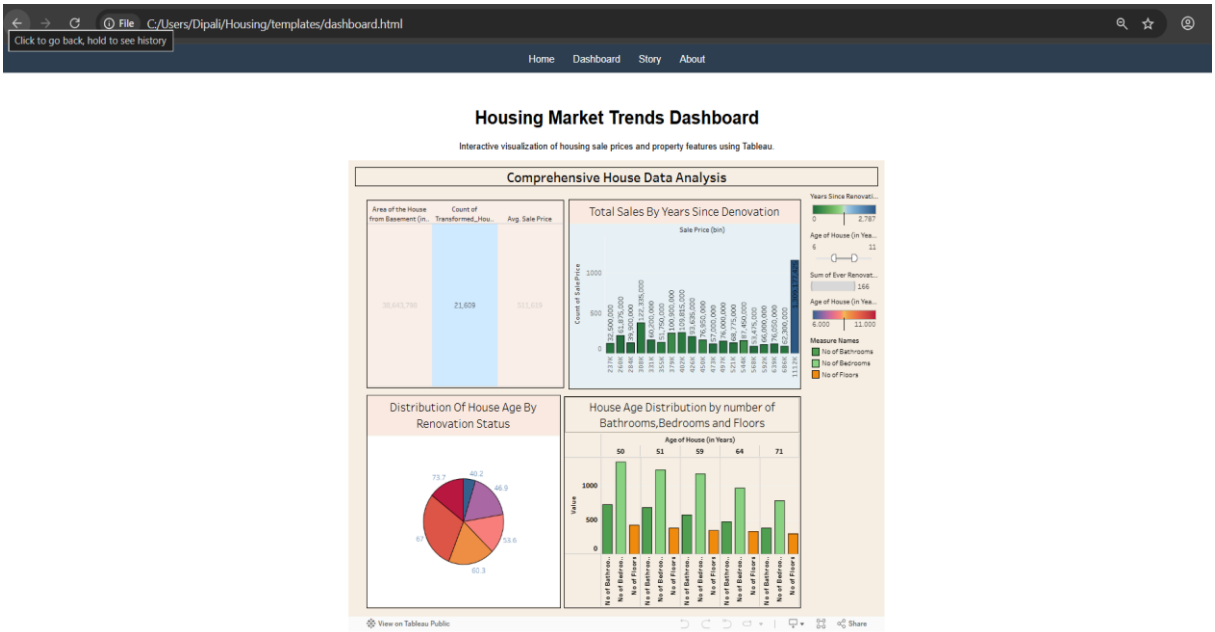
Visualizing Housing Market Trends
An Analysis of Sale Prices and Features using Tableau. This project explores housing market data using interactive dashboards to understand trends in property prices and housing features.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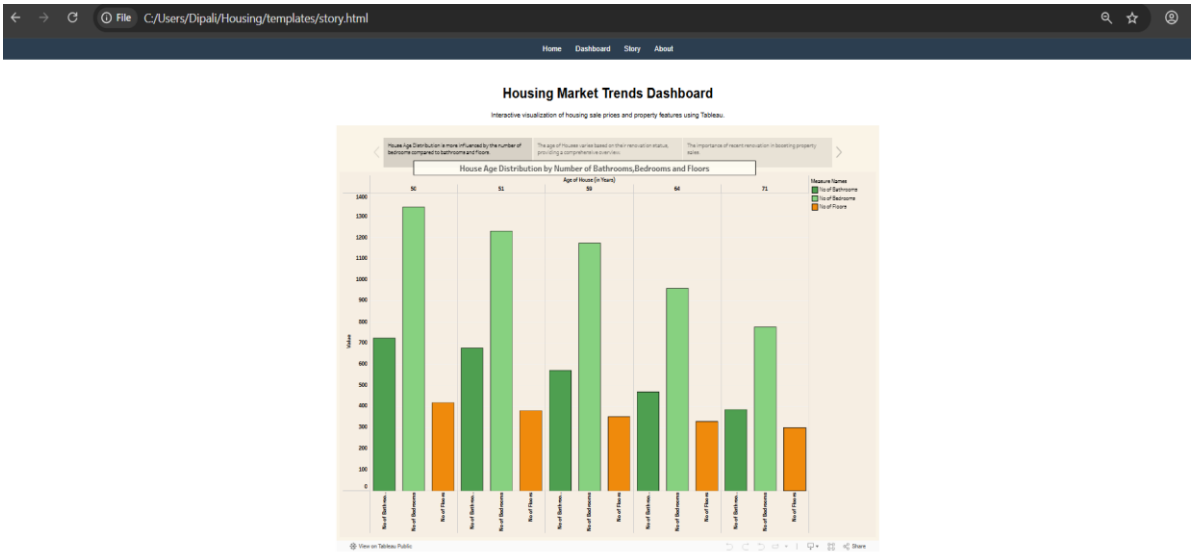
Click to go back, hold to see history
Home Dashboard Story About
Housing Market Trends Dashboard
Interactive visualization of housing sale prices and property features using Tableau.
Comprehensive House Data Analysis
Total Sales By Years Since Denovation
Distribution Of House Age By Renovation Status
House Age Distribution by number of Bathrooms,Bedrooms and Floors
View on Tableau Public
Share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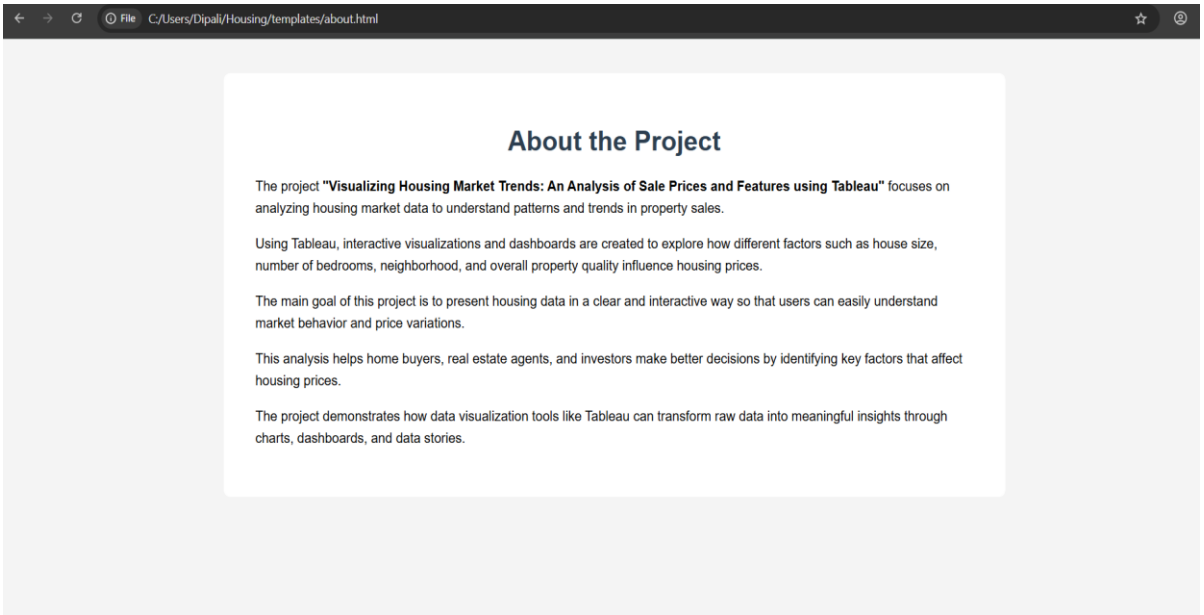
Home Dashboard Story About
Housing Market Trends Dashboard
Interactive visualization of housing sale prices and property features using Tableau.
House Age Distribution by Number of Bathrooms,Bedrooms and Floors
Age of House (in Years)
50
51
59
64
71
Measure Names
No of Bathrooms
No of Bedrooms
No of Floors
Values
View on Tableau Public
Share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About the Project
The project **"Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau"** focuses on analyzing housing market data to understand patterns and trends in property sales.
Using Tableau, interactive visualizations and dashboards are created to explore how different factors such as house size, number of bedrooms, neighborhood, and overall property quality influence housing prices.
The main goal of this project is to present housing data in a clear and interactive way so that users can easily understand market behavior and price variations.
This analysis helps home buyers, real estate agents, and investors make better decisions by identifying key factors that affect housing prices.
The project demonstrates how data visualization tools like Tableau can transform raw data into meaningful insights through charts, dashboards, and data stories.

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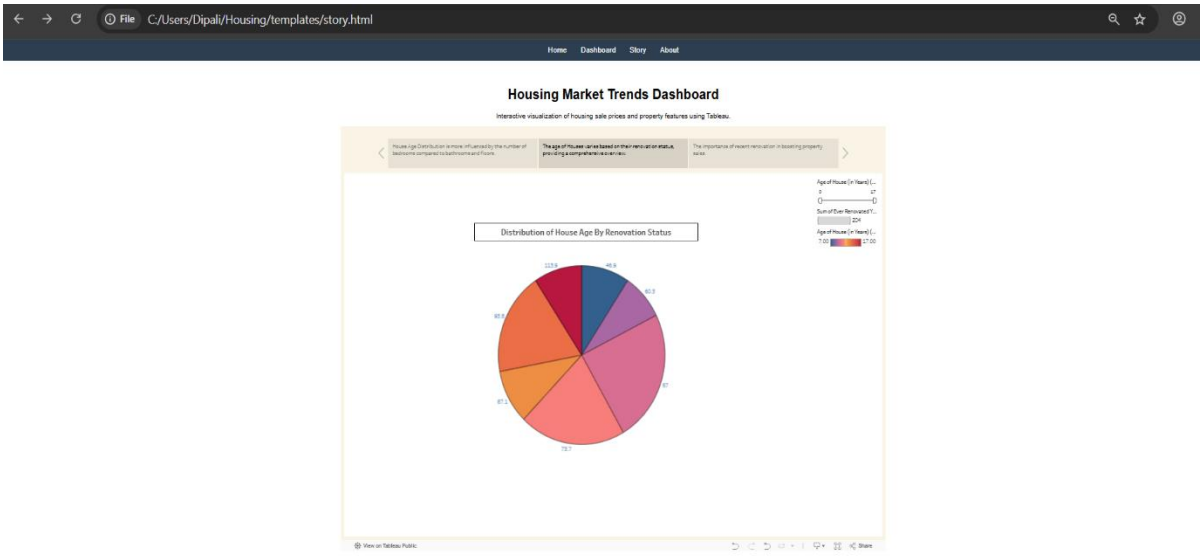
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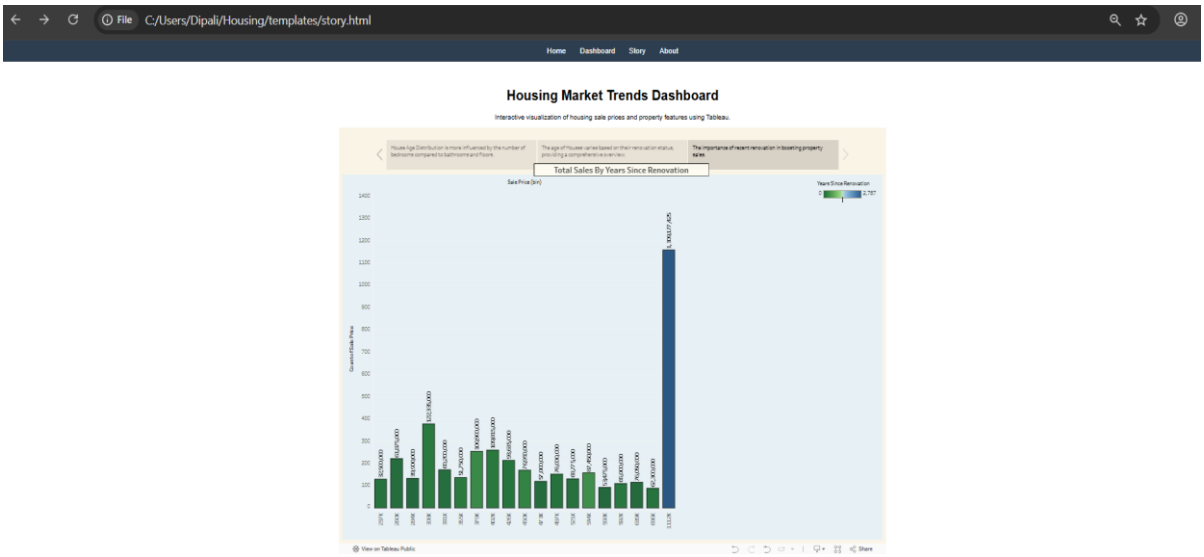
Housing Market Trends Dashboard
Interactive visualization of housing sale prices and property features using Tableau.
Distribution of House Age By Renovation Status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